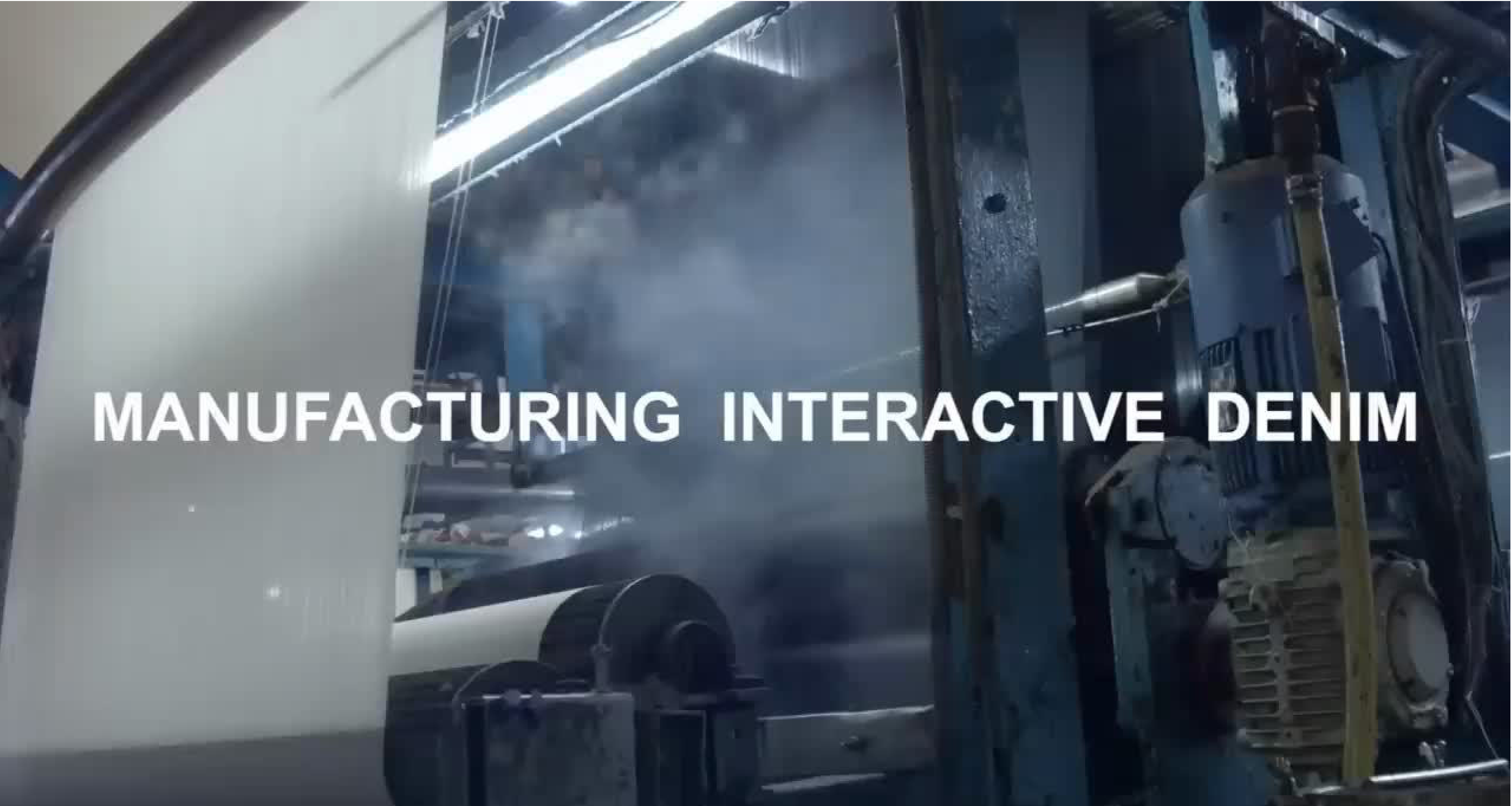
E-Textiles and Functional Materials

slides by Dr. Marion Koelle
presented by Anika Bork

inf175
University of Oldenburg, Summer Term 2024

1. **The Musical Jacket (1998)**
2. www.maggieorth.com/art_Jacket.html
3. Orth, M., Smith, J. R., Post, E. R., Strickon, J. A., & Cooper, E. B. (1998). Musical Jacket. In ACM SIGGRAPH 98 Electronic Art and Animation Catalog (p. 38); online at https://digitalartarchive.siggraph.org/artwork/maggie-orth-j-r-smith-e-r-post-j-a-strickon-e-b-cooper-musical-jacket/
4. Video: Joe Paradiso, https://paradiso.media.mit.edu/SpectrumWeb/captions/Jackets.html



1. Google ATAP, Project Jaquard: Making the Jacket, online at https://www.youtube.com/watch?v=grKHwQIaiaA (retrieved 04/2024)

Hannah Perner-Wilson and Mika Satomi, Kobakant (2017), https://www.kobakant.at/DIY/?p=6745

Learning Goals

1. After this lecture you can …
2. name common types of conductive (e-textile) materials.
3. interpret data sheets describing conductive (e-textile) materials.
4. determine whether a given material is electrically conductive
5. determine the resistivity of a given material.
6. make an informed selection of an e-textile material to match your project‘s requirements.

Legend:

- Recommended reading
- References
- Solo Exercise
- Group Exercise / Discussion
- Hands-on Exercise

E-Textile Sensors and other Components

Contact Sensors (Digital)

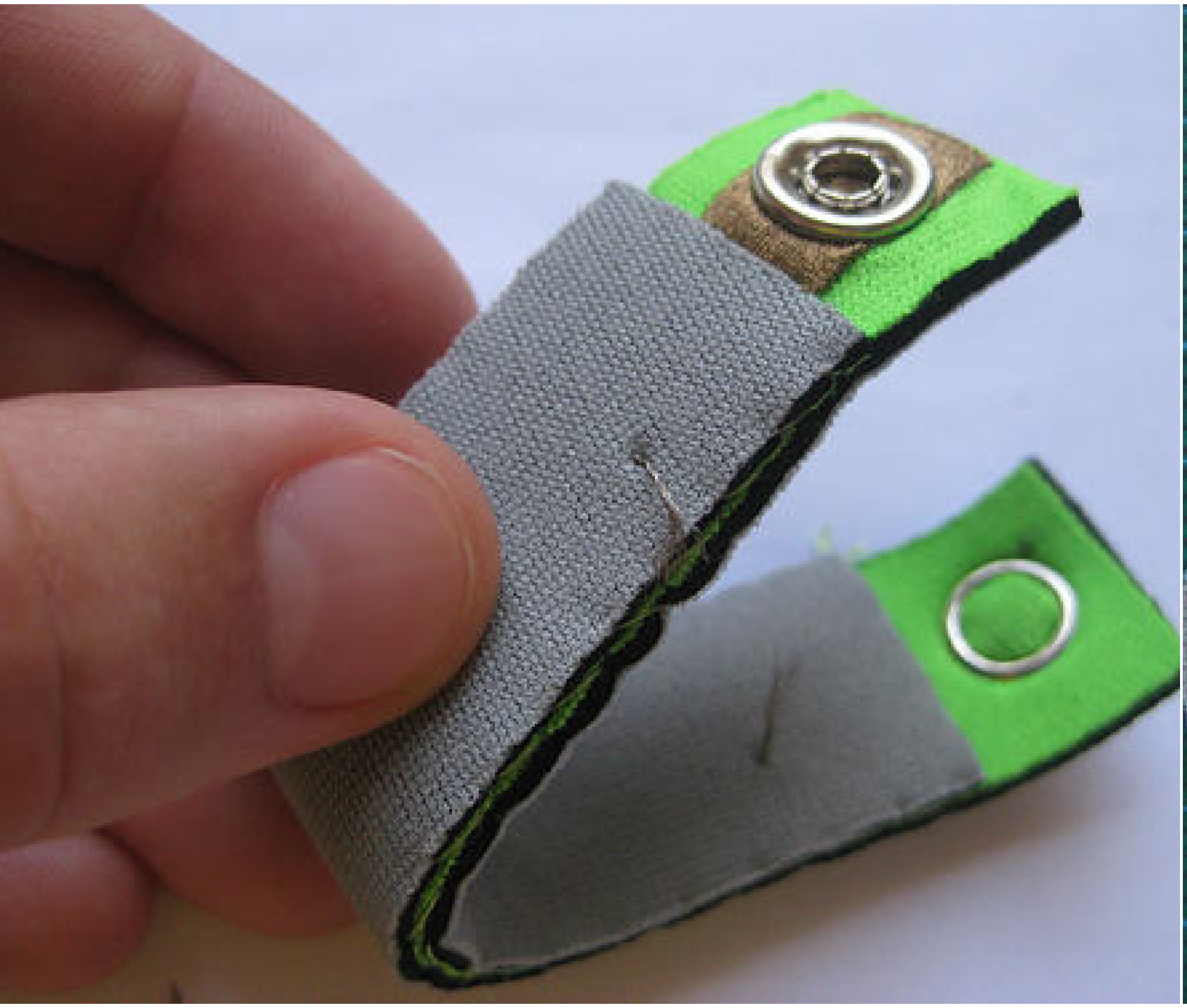
Fabric Button
http://www.kobakant.at/DIY/?p=48

Tilt Sensor
http://www.kobakant.at/DIY/?p=201

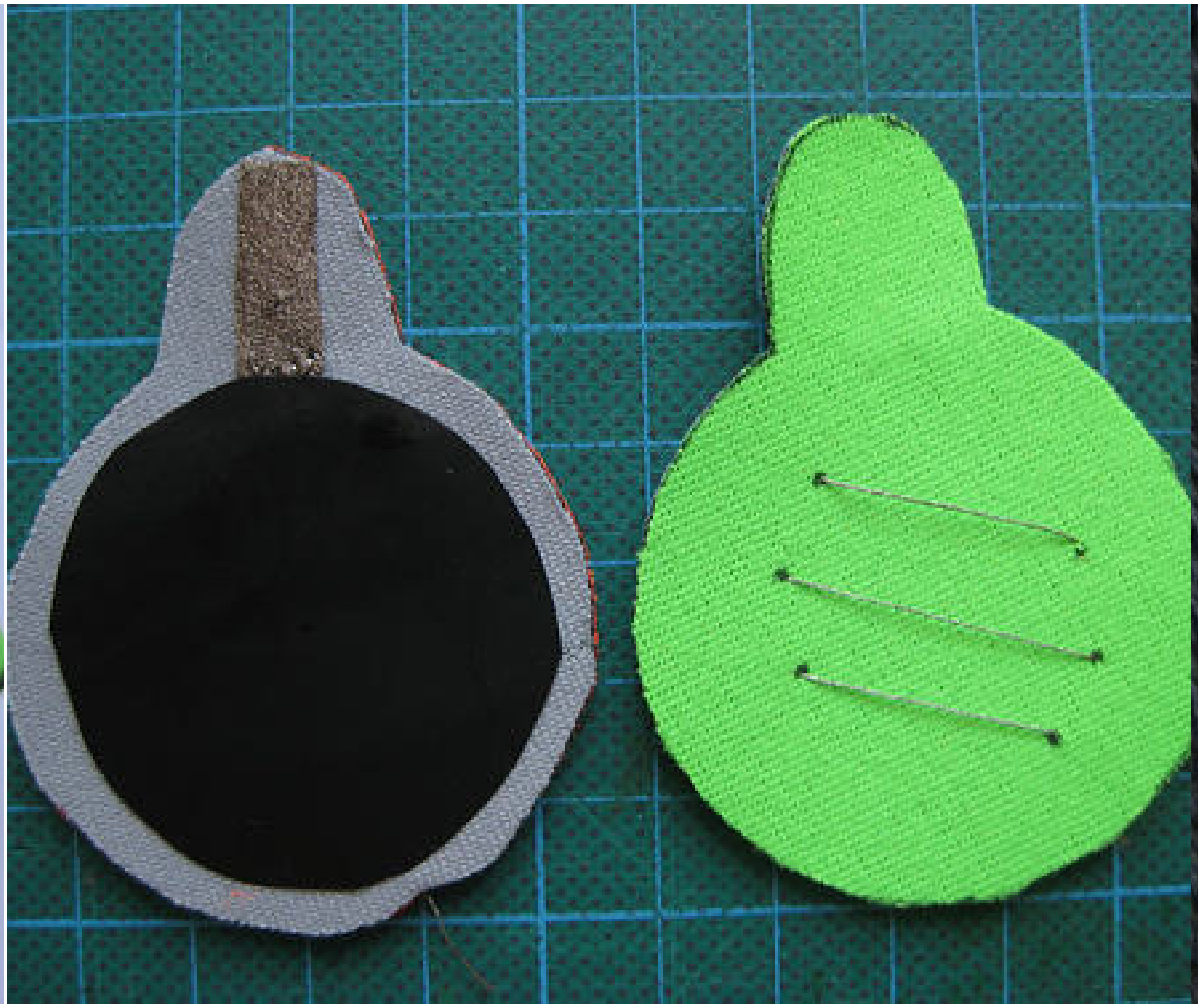
Neoprene stroke bracelet
http://www.kobakant.at/DIY/?p=5807

1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online a https://www.kobakant.at (retrieved o4/2024)

Stretch and Pressure Sensors (Analog)



Neoprene Bend Sensor
http://www.kobakant.at/DIY/?p=20



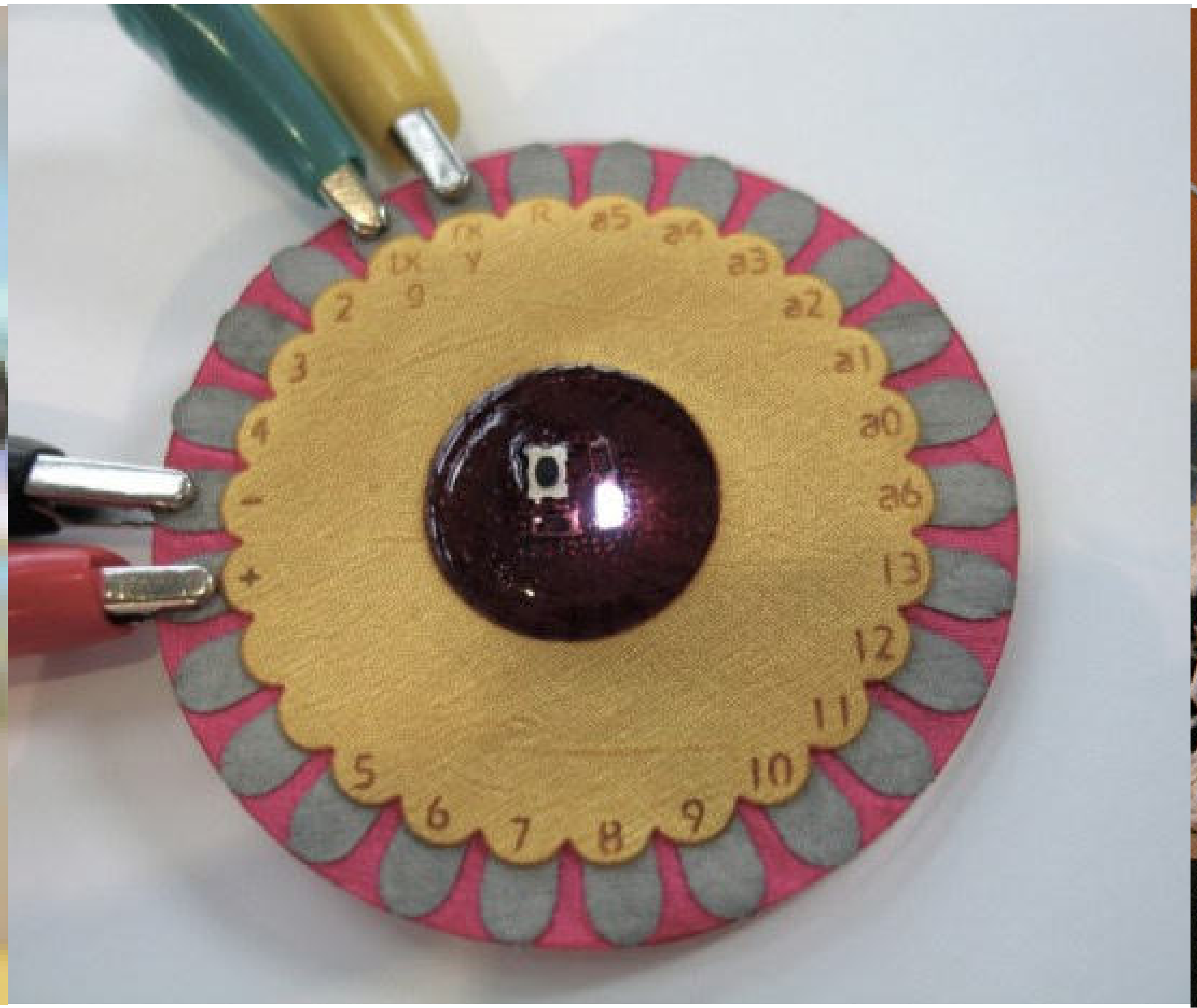
Neoprene Pressure Sensor
http://www.kobakant.at/DIY/?p=65

Knit Stretch Sensor
http://www.kobakant.at/DIY/?p=2108

1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online a https://www.kobakant.at (retrieved o4/2024)

Textile Microcontroller Boards

Fabric Breakout Board for Teensy
https://www.kobakant.at/DIY/?p=2672

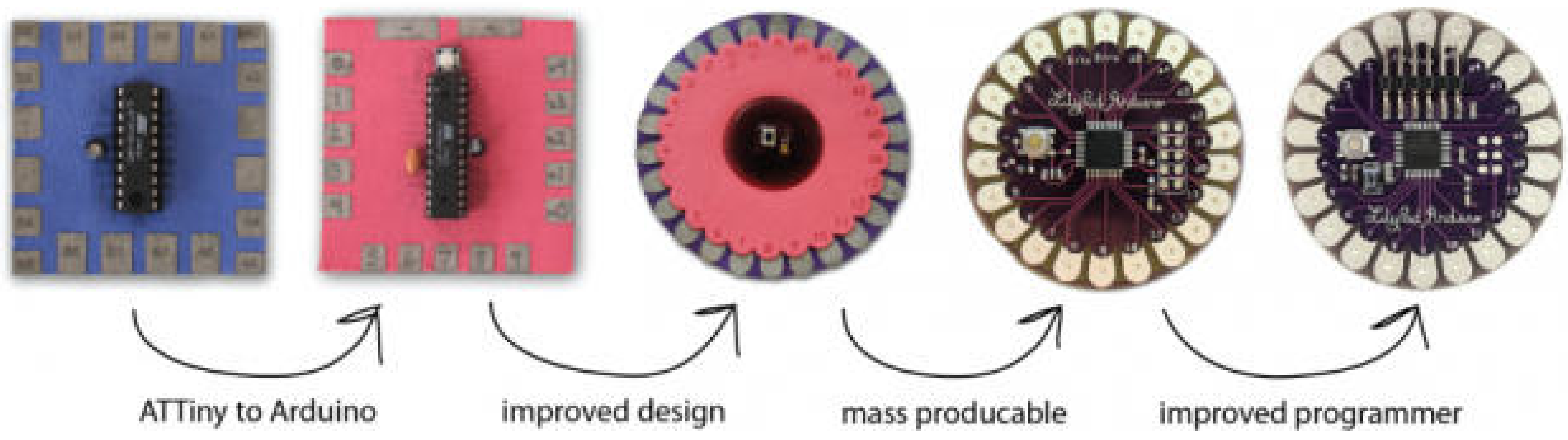


Textile Lilypad Board, Leah Buechley
Makezine (2014)

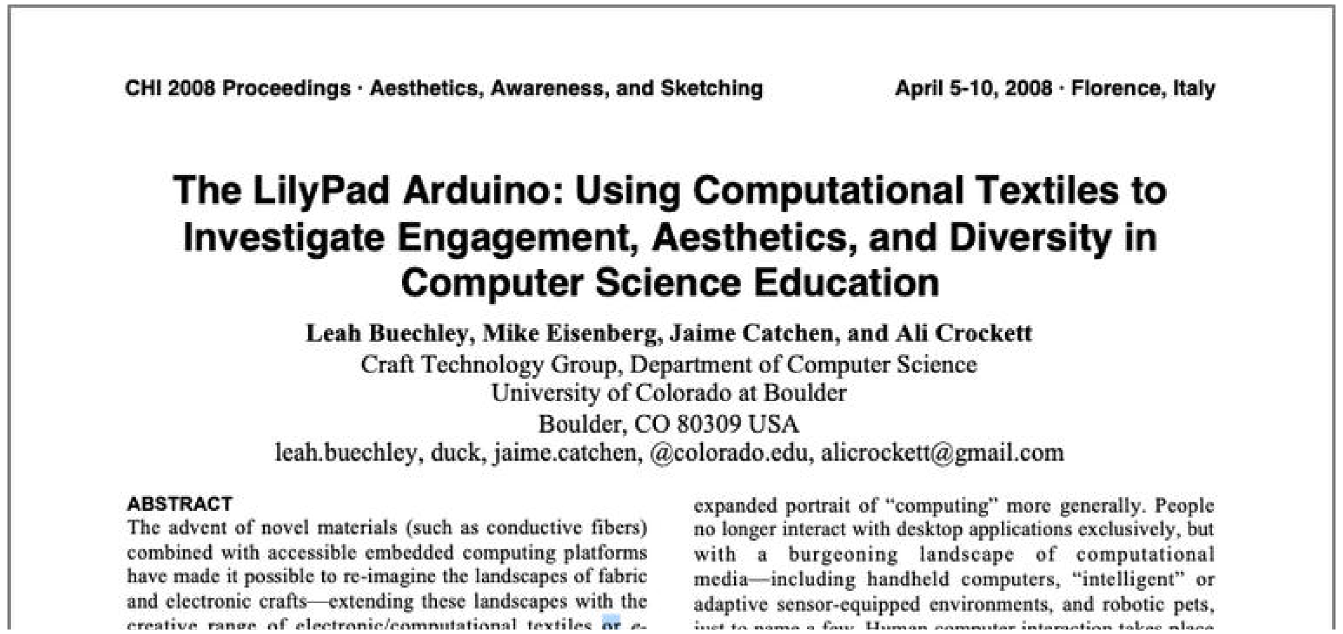
Embroidered Lilypad Circuit, Becky Stern
Wikimedia Commons, CC-BY-SA 2.0

1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online a https://www.kobakant.at (retrieved o4/2024)
   Leah Buechley: Crafting the Lilypad Arduino (2014), online at https://makezine.com/article/maker-news/leah-buechley-crafting-the-lilypad-arduino/ (retrieved 04/2024)

Arduino LilyPad



Textile Lilypad Board, Leah Buechley
Makezine (2014)



CHI 2008 Proceedings · Aesthetics, Awareness, and Sketching — April 5-10, 2008 · Florence, Italy

The LilyPad Arduino: Using Computational Textiles to Investigate Engagement, Aesthetics, and Diversity in Computer Science Education

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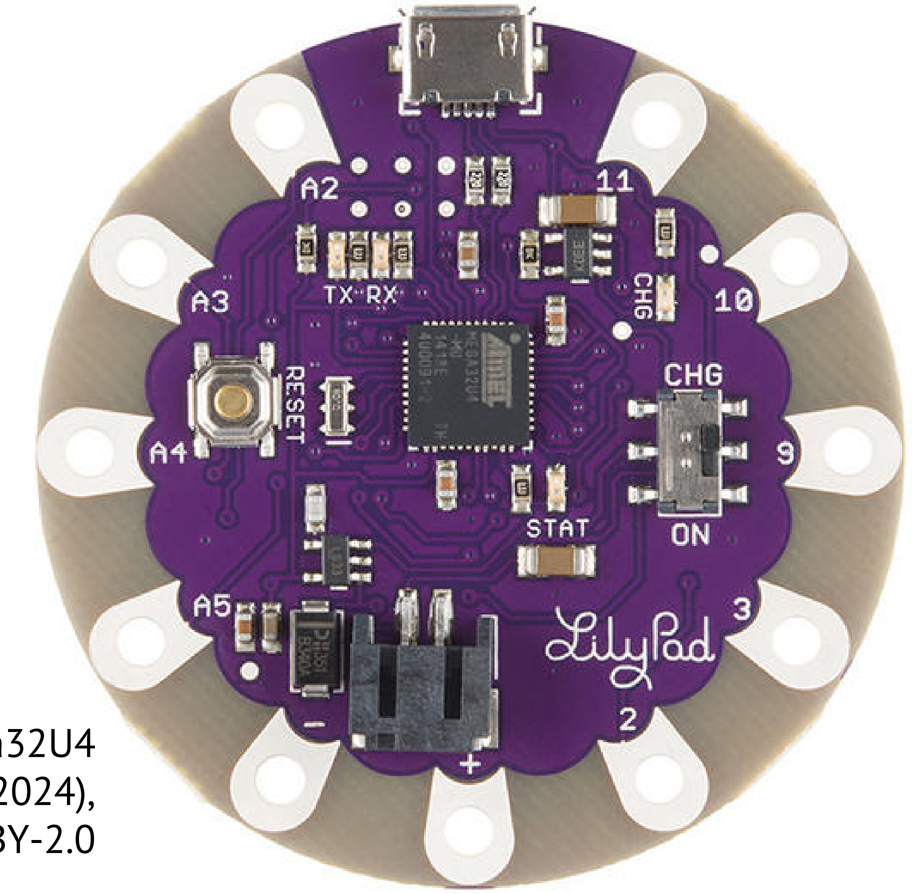
ABSTRACT
The advent of novel materials (such as conductive fibers) combined with accessible embedded computing platforms have made it possible to re-imagine the landscapes of fabric and electronic crafts—extending these landscapes with the

expanded portrait of "computing" more generally. People no longer interact with desktop applications exclusively, but with a burgeoning landscape of computational media—including handheld computers, "intelligent" or adaptive sensor-equipped environments, and robotic pets,

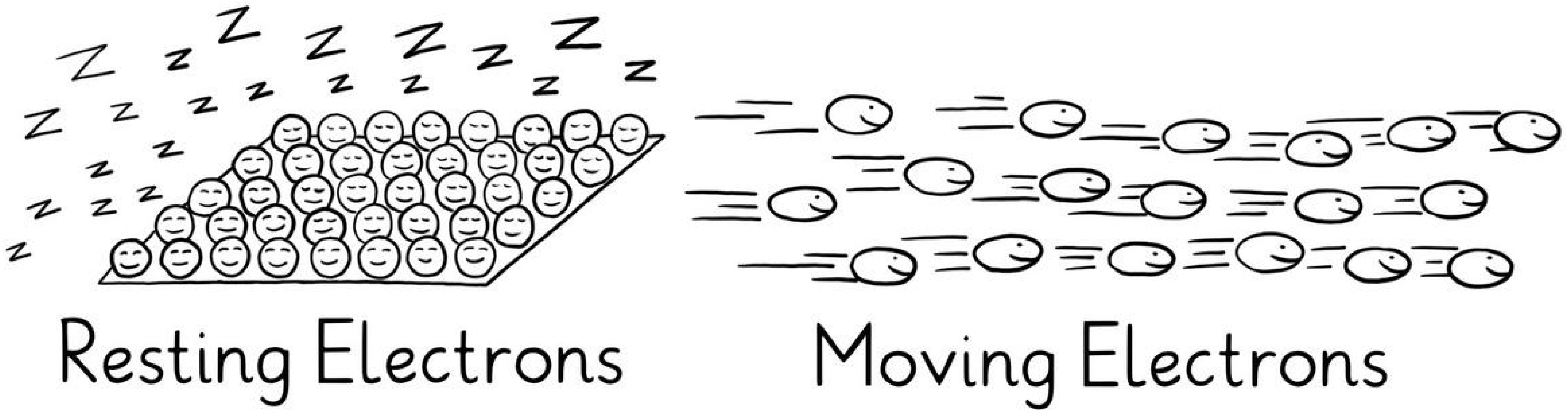


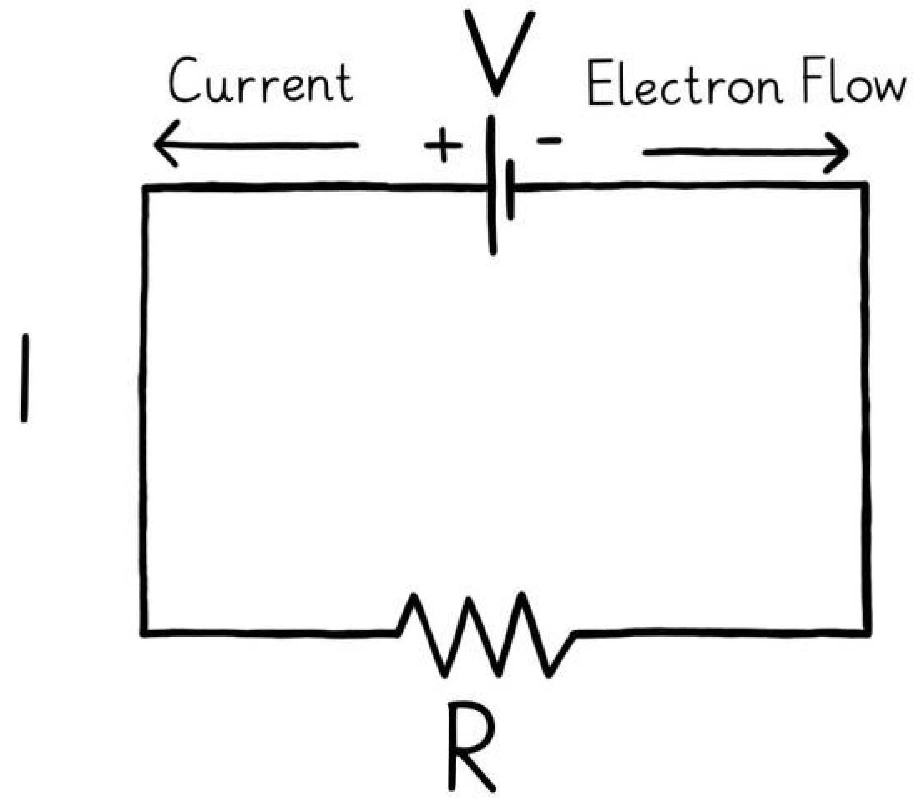
Arduino LilyPad USB – ATmega32U4
Board (produced by Sparkfun, 2024),
Image by Sparkfun, CC-BY-2.0

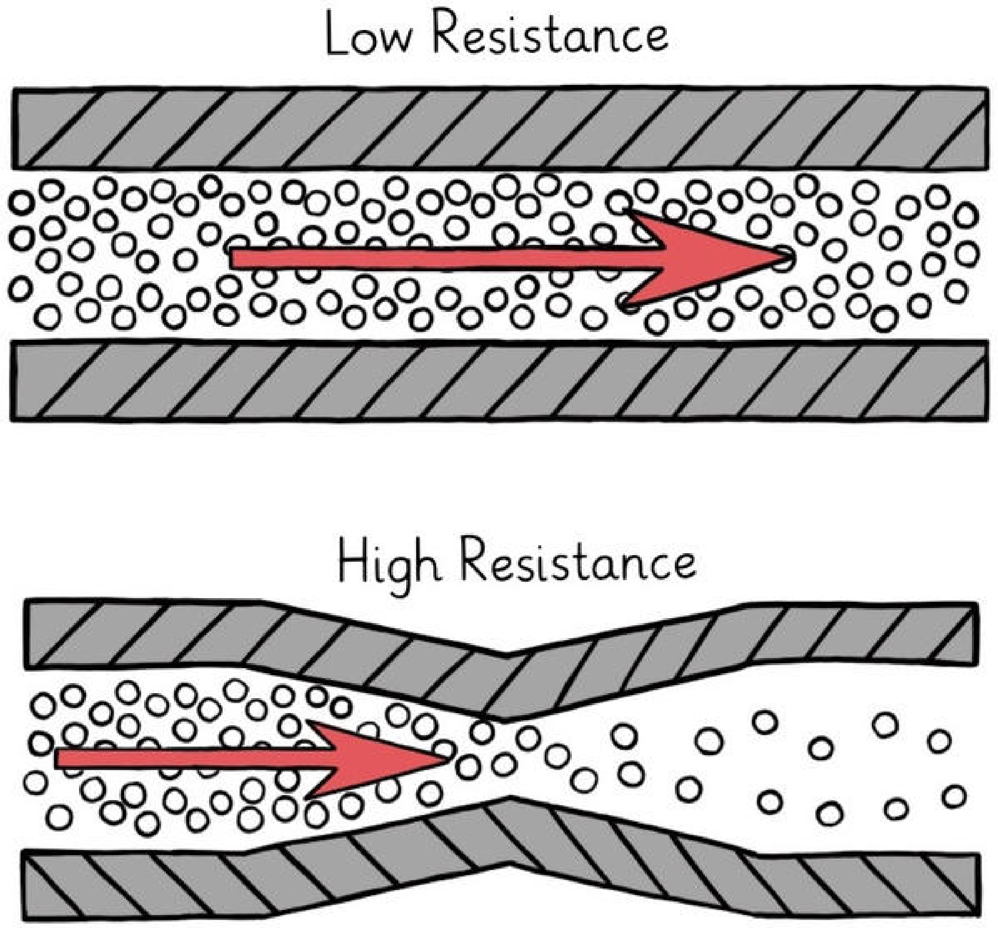
1. Leah Buechley: Crafting the Lilypad Arduino (2014), online at https://makezine.com/article/maker-news/leah-buechley-crafting-the-lilypad-arduino/ (retrieved 04/2024)
Buechley, L., Eisenberg, M., Catchen, J., & Crockett, A. (2008, April). The LilyPad Arduino: Using Computational Textiles to Investigate Engagement, Aesthetics, and Diversity in Computer Science Education. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (pp. 423-432).

Conductivity & Resistivity

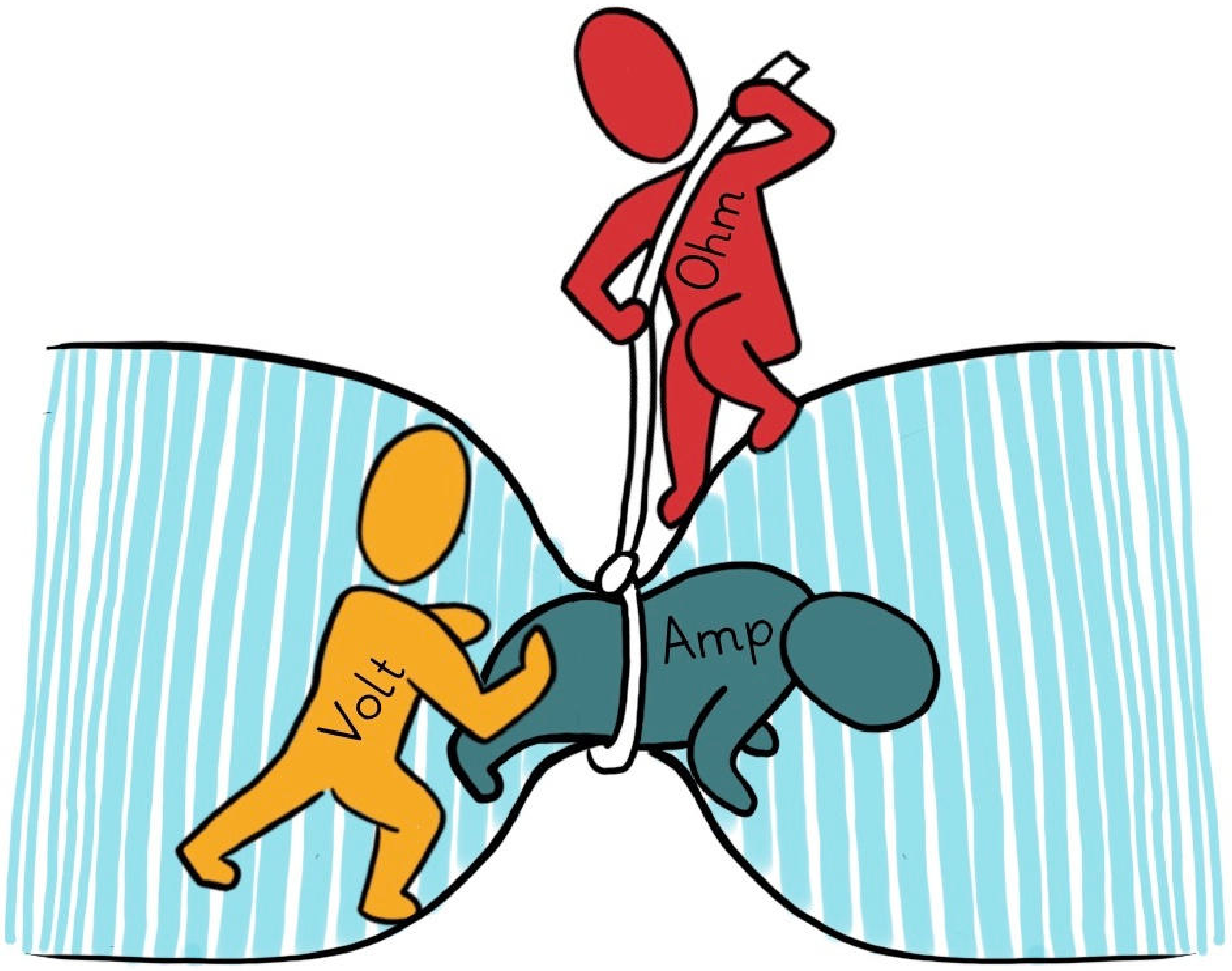
Electrons & Conductivity







Ohm's Law



1. = Voltage (like a battery, expressed in Volt)
2. = Current (electron flow, expressed in Ampere, A colloquially also “amps”)
3. = Resistance (extend to which the material or component resists electron flow, expressed in ohms, Ω)

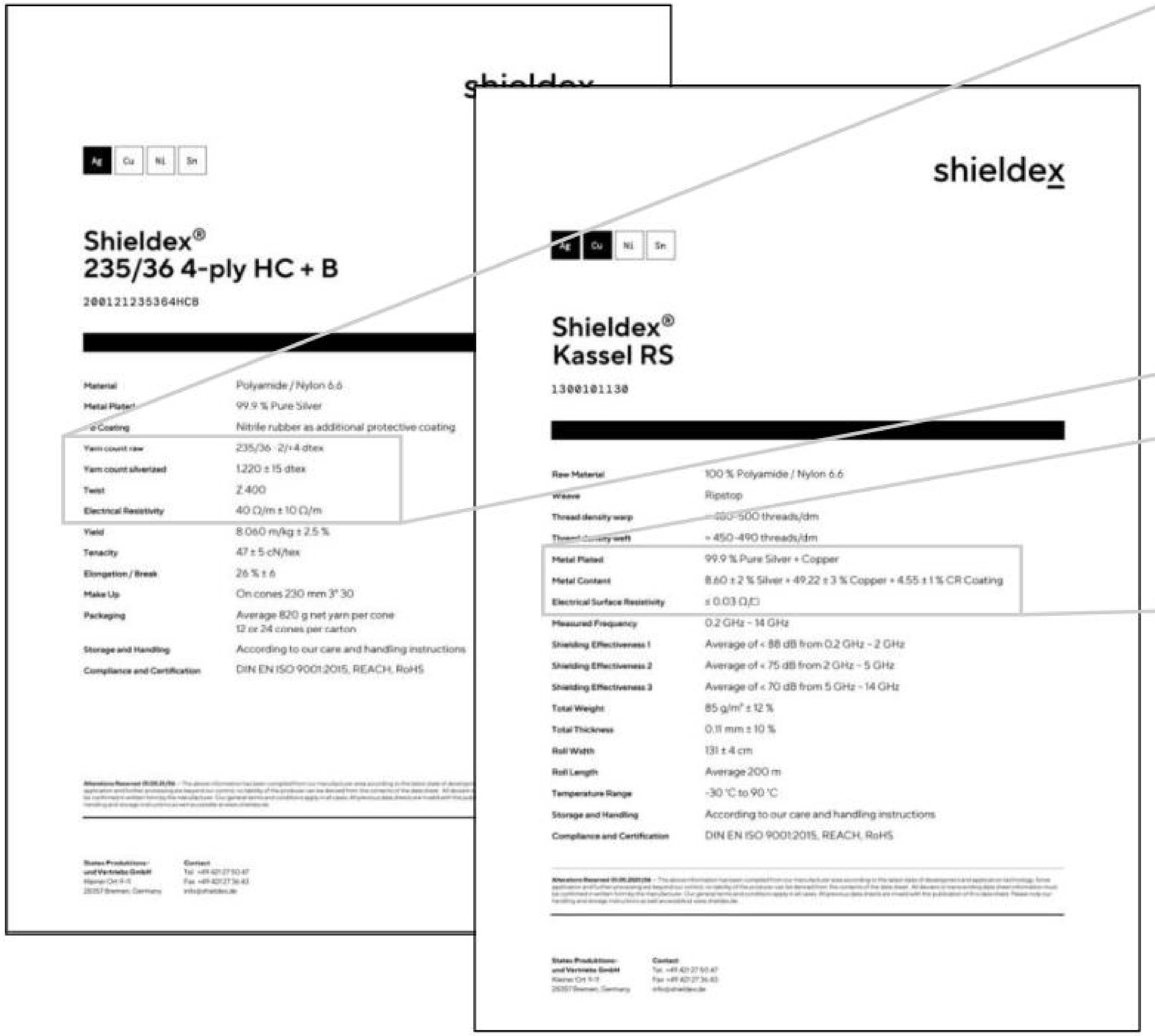
Using a Multimeter

1. Different measurement modes:
   2. Resistance
   3. Current
   4. Voltage

   5. Crucial tool to diagnose circuits, testing batteries, and determining a materials conductivity.

1. Tutorial: How to use a Multimeter, Sparkfun (2015), https://learn.sparkfun.com/tutorials/how-to-use-a-multimeter/
2. Image Source: Voelkner, Voltcraft MT-52-SE

Reading Data Sheets – Conductivity



Yarn count raw	235/36 -2/+4 dtex
Yarn count silverized	1.220 ± 15 dtex
Twist	Z 400
Electrical Resistivity	40 Ω/m ± 10 Ω/m

Metal Plated	99.9 % Pure Silver + Copper
Metal Content	8.60 ± 2 % Silver + 49.22 ± 3 % Copper + 4.55 ± 1 % CR Coating
Electrical Surface Resistivity	≤ 0.03 Ω/□

Pay attention!: for e-textile materials, resistivity is typically not given in ohms (Ω).

- Volume resistivity: Ω/m (ohm-meter), used for conductive thread or yarn.
- Sheet resistivity: Ω/☐ (ohm-square), used for conductive fabric, films or traces.

Volume Resistivity (Ω/cm or Ω/m)

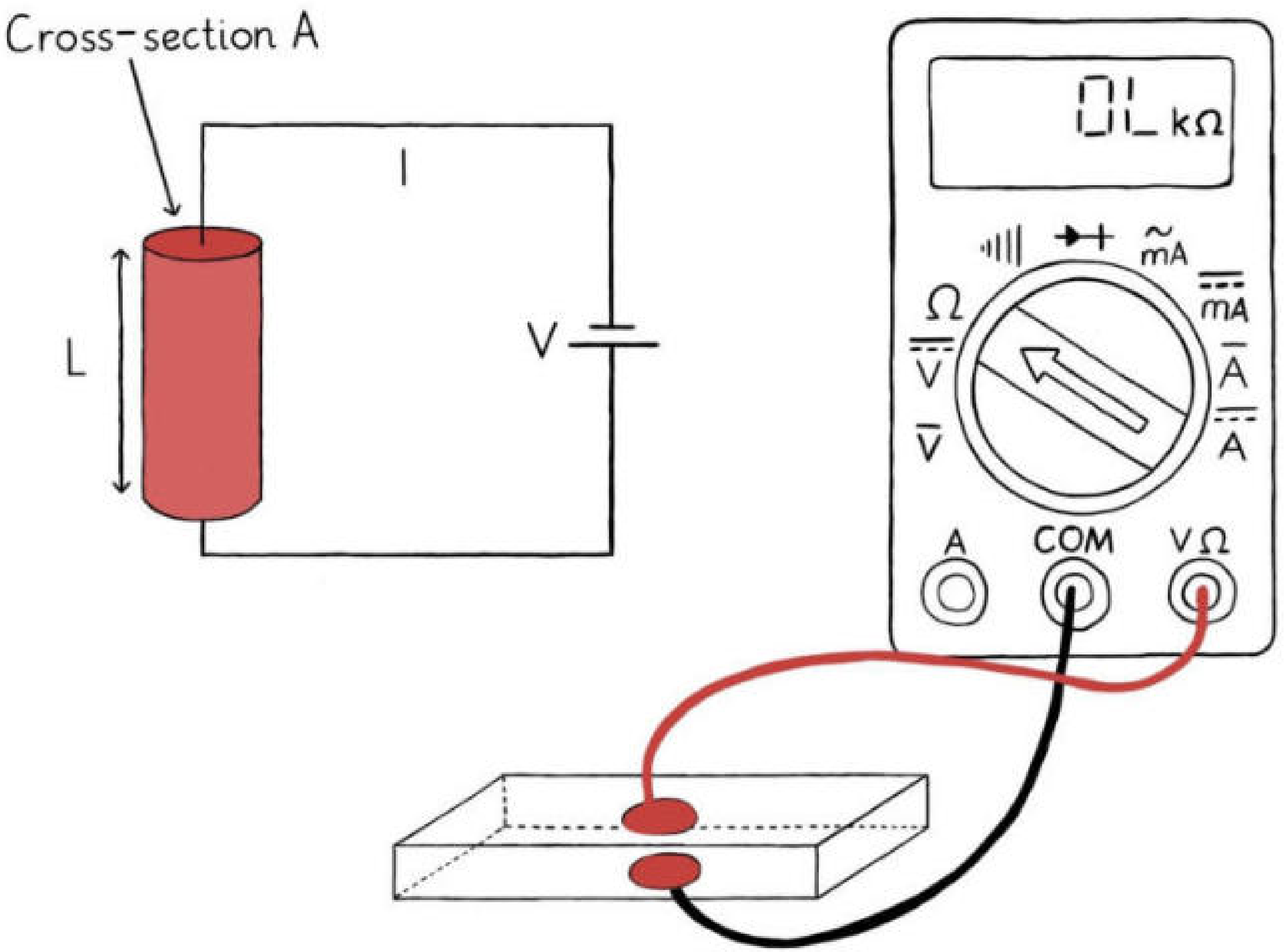
$$p = R * A/L$$

p = Volume Resistivity (Ω/cm)

R = Resistance (Ω)

A = Cross-sectional area of the material (cm^2), can be diameter (e.g., of a yarn) or width*thickness (e.g., of a foam).

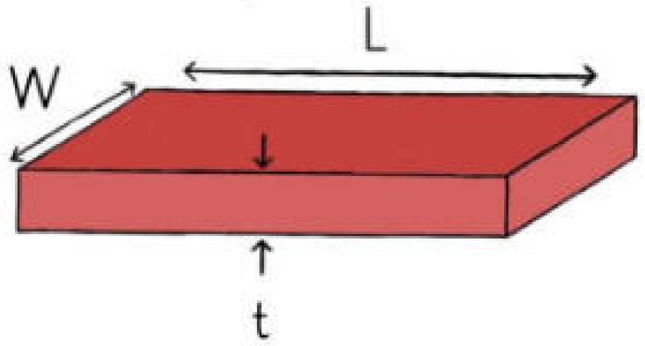
L = Distance between the probe electrodes (cm).



Volume resistivity (or bulk resistivity) is used to measure how strongly a material resists the flow of electric current. It is expressed in Ω/cm.

Surface Resistivity (Ω/sq or Ω/☐)

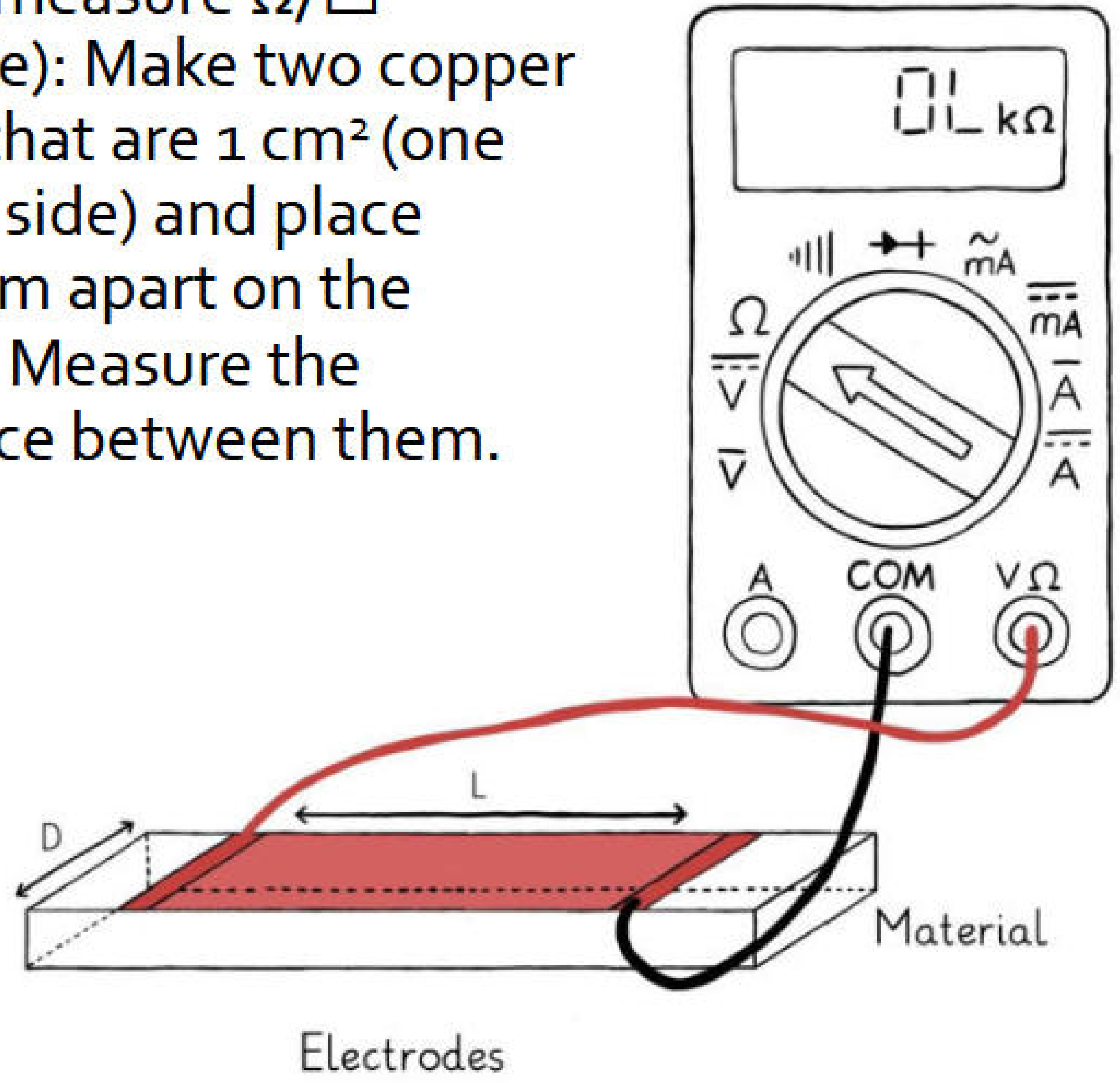
Surface resistivity (also sheet resistivity) describes the resistance of a **square piece** of a **thin** material with contacts made to two opposite sides of the square.



Applies to materials where thickness *t* is negligible, e.g., thin films, or fabrics.

Expressed in Ω/sq or Ω/☐; spoken „ohms-per-square"

How to measure Ω/☐ (example): Make two copper probes that are 1 cm^2 (one cm on a side) and place them 1cm apart on the surface. Measure the resistance between them.



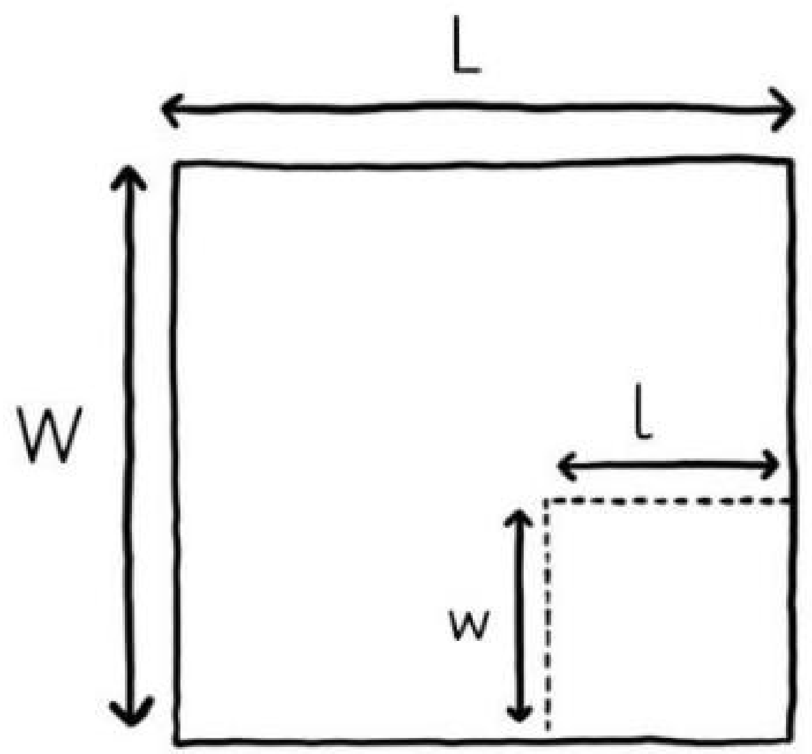
Surface Resistivity (Ω/sq or Ω/☐)

Important: surface resistivity is <u>dimensionless</u>, i.e., it does not change as long as the measurement is performed <u>correctly</u> using a probe with square dimensions. Great for making comparisons!

$$p_s = \frac{p_v}{t} = \frac{W}{L} * \mathrm{R}$$

$$if\ L = W;\ l = w:$$

$$p_s = R\ [\Omega/\square]$$



p_s = Surface Resistivity (Ω/☐)

p_v = Volume Resistivity (Ω/cm)

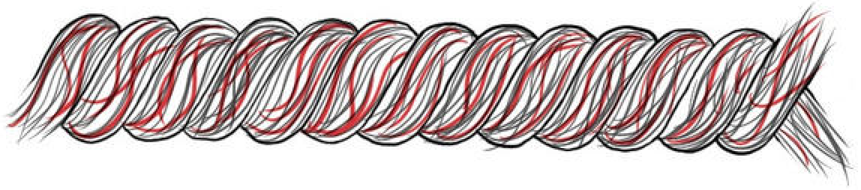
R = Resistance (Ω)

W = Width of electrodes

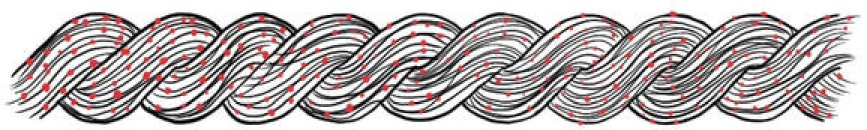
L = Distance between the probe electrodes (cm).

Conductive Threads, Yarns and Fibers

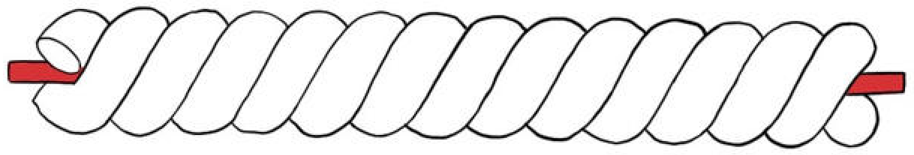
Different Types of Conductive Yarns



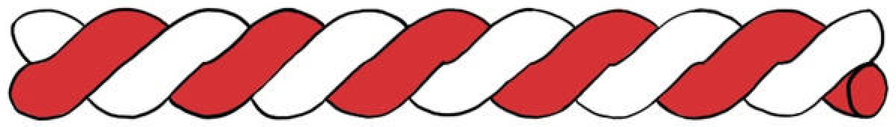
Spun from a mix of conductive and non-conductive filaments



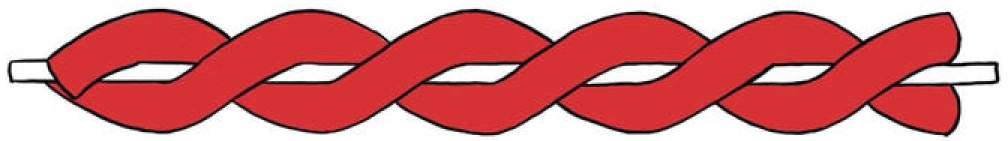
Spun from non-conductive fibers or filaments, then coated in a thin layer of metal (metallized) or conductive polymer (polymerized)



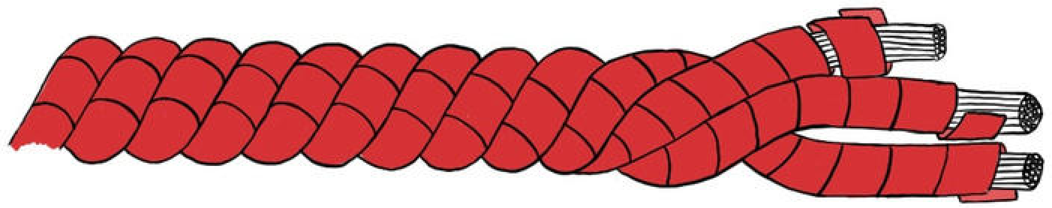
Metal-filled yarn (e.g., containing a stainless steel core)



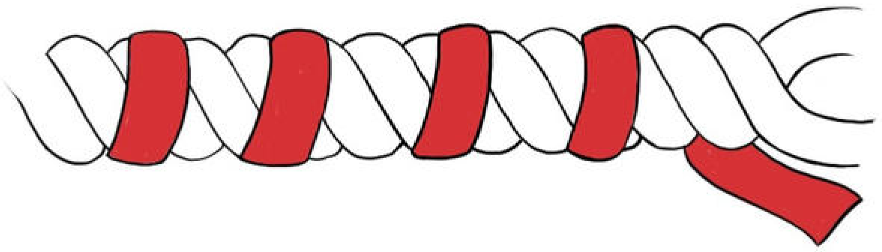
Conductive and non-conductive yarns plied together



Metal wrapped yarn



Filaments (e.g., nylon) wrapped in metal, then spun together



Spun fibers wrapped in metal (e.g., so-called gimp yarn)

Stainless Steel Thread

1. Stainless steel fibers spun together
2. Thickness and no of ply can vary, commonly found are 2ply, 3ply and (sometimes) 4ply-
3. Resistance: ca. 8.5 ohm/m (28 ohm/ft), i.e., **moderately conductive.**
4. Not (well) machine sewable (!), not solderable, rather “hairy” but can be waxed to be easier to sew.
5. Relatively wide-spread, e.g.,
   a. https://www.sparkfun.com/products/11791
   b. https://www.sparkfun.com/products/13814

Example Material

2ply

4ply

6ply

Metal-plated Thread

1. Silver-plated Nylon, optionally coated with e.g., Nitrile Rubber
2. Resistance: between 30 ohm/m and 80 ohm/m depending on ply, i.e., **moderately conductive.**
3. Smooth, machine sewable, not (!) solderable.
4. Tarnishes easily due to the silver (esp. when sewn by hand), coating may help.
5. Available as, e.g.,
   a. Shieldex 235/36 x6 HCB (6ply, <30 Ω/m)
   b. Shieldex 235/36 x4 HCB (4ply , <40 Ω/m)
   c. Shieldex 235/36 x2 HCB (2ply, <80 Ω/m)
   d. Sparkfun Conductive Thread 117/17 2ply (legacy, still available in the lab)
   e. Sparkfun Conductive Thread 234/34 4ply (legacy, still available in the lab)

Metal-wrapped Nylon Thread

1. Thin flattened wires wrapped around a nylon core, then spun together (so called “Kupfer- or Silbergespinst, Lahnspinnung”)
2. Resistance: > 2 Ohm/m – i.e., **highly conductive**, e.g., for embroidered speakers
3. Still very flexible but slightly stiffer than metallized yarns, machine sewable.
4. Solderable!
5. Available as (e.g.) “Karl Grimm High Flex Kupfer 7x1” to business customers in larg(er) quantities
6. More information: https://www.kobakant.at/DIY/?p=379

Example Material

“Data Stretch“

1. Thin, flat copper wire wound as spiral around a flexible core, then wrapped with a stretchable textile cover.
2. Resistance: 5 Ohm/m – i.e., **highly conductive**, suitable e.g., for multitouch sensing.
3. Insulated due to the textile outer layer
4. Stretchable, about 30% elongation.
5. Requires a lot of practice to solder.
6. Available as (e.g.)
   a. TibTech Data Stretch

Example Material

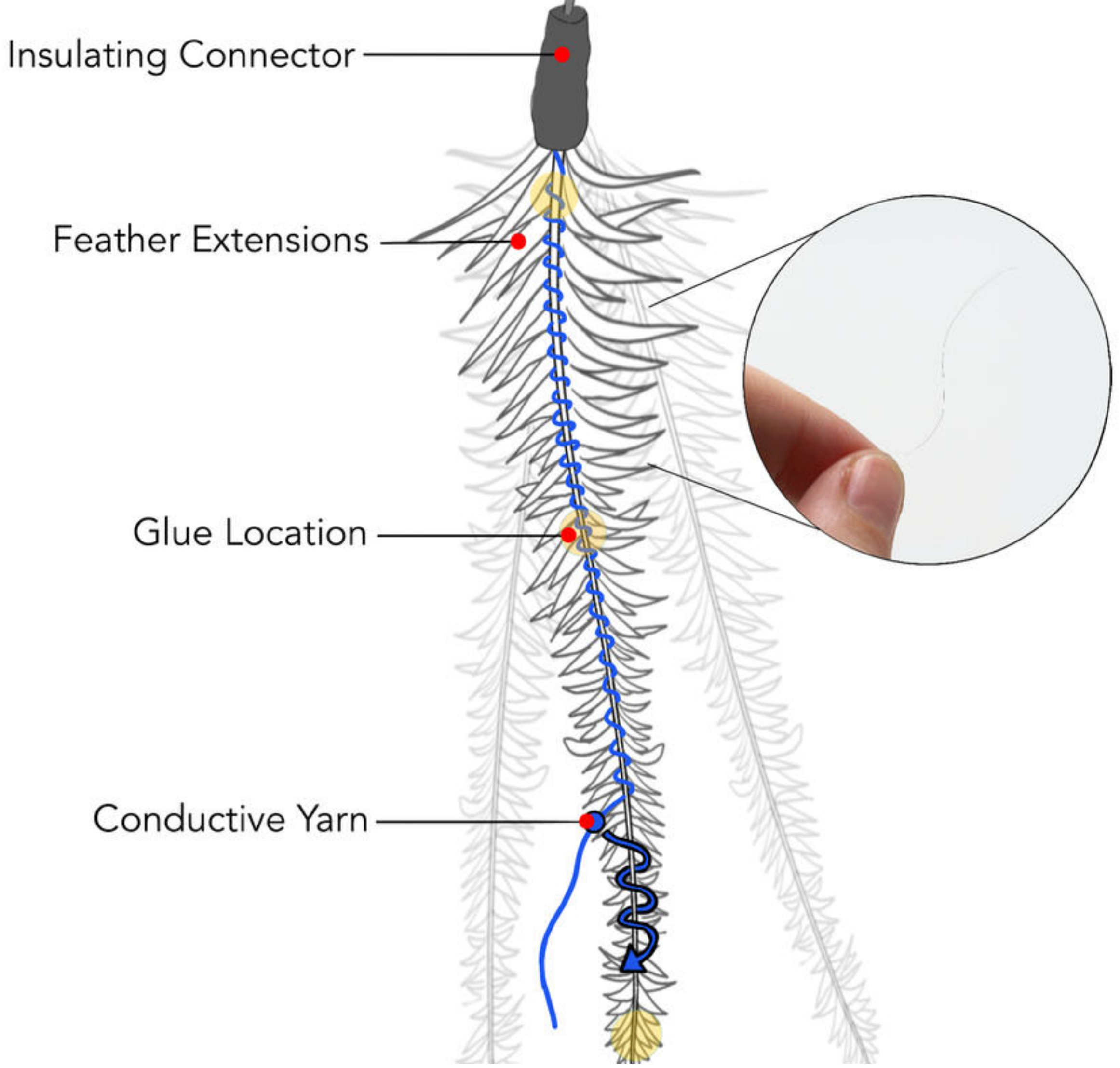
Conductive Wool

1. Conductive fibers (e.g., stainless steel) mixed in with (e.g.) merino wool.
2. Usually **moderately conductive**, can be shaped into sensors with **variable resistance**: movement changes resistance between fibers.
3. Can be used as a basis for felted sensors (e.g.,) or hand-spun into yarn and then knit, crocheted or braided.
4. Not solderable.
5. Available e.g., as
   a. Bekinox by Baekert
   b. Barts & Francis Conductive Wool

Example Material

Metal Filaments

1. Pure metal (e.g., stainless steel), extruded extremely thin (e.g., <0.04mm)
2. Usually **highly conductive.**
3. Often used as a basis for compound yarns or mixed with wool, but also available as plain filaments.
4. Extremely thin, thus hard to handle, sew or solder.
5. Available e.g., as
   a. https://www.adafruit.com/product/1088
   b. Barts & Francis 0.355mm Inox
   c. Barts & Francis twisted Inox



1. **Featherhair (DIS 2022)**
2. Marie Muelhaus, Jürgen Steimle, and Marion Koelle
3. https://github.com/HCI-Lab-Saarland/FeatherHair

Metal filled „Craft“

1. Fibers plied together or filled with a metal filament (often stainless steel).
2. Usually **highly conductive** but no data sheet (!)
3. Solderable but requires practice due to the filament’s thin diameter.
4. Comes in a large variety of colors and materials, e.g., silk, cotton, merino, …
5. Common brands/suppliers:
    a. ITO
    b. Habu
    c. Barts&Francis

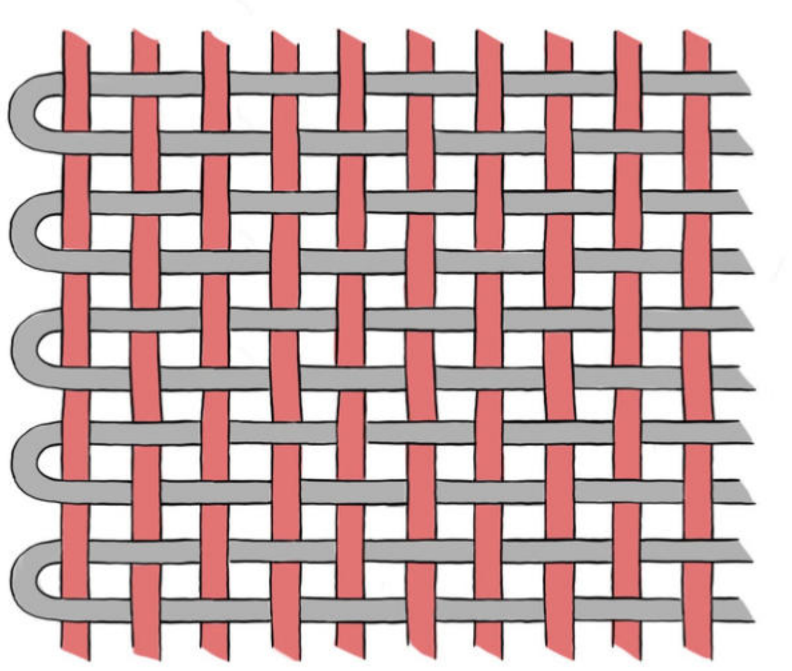
Example Material

Jewelry Wire

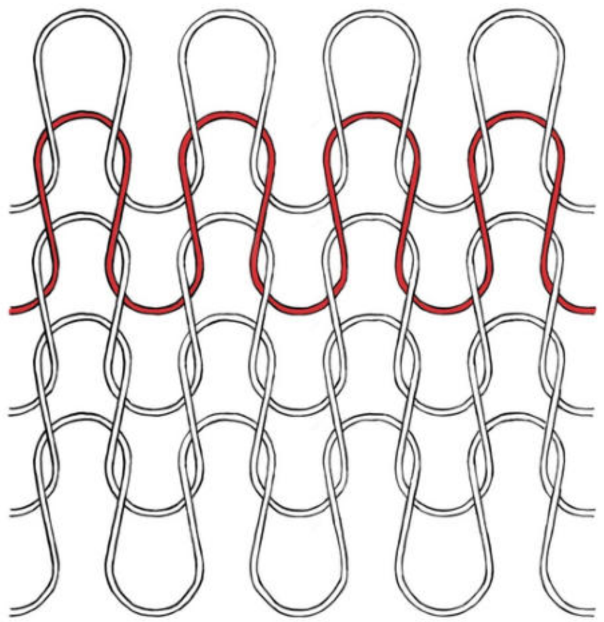
1. Often made from silver, copper, zinc or combinations thereof (e.g., brass).
2. Usually **highly conductive** but not data sheet (!)
3. Solderable (easy).
4. Can be blank or coated (e.g., with nylon).
5. Usually considerably thicker than metal filaments.
6. Rather stiff compared to conductive threads or yarns, not sewable
7. Easy to obtain, e.g., from craft supply stores.

Conductive Fabrics

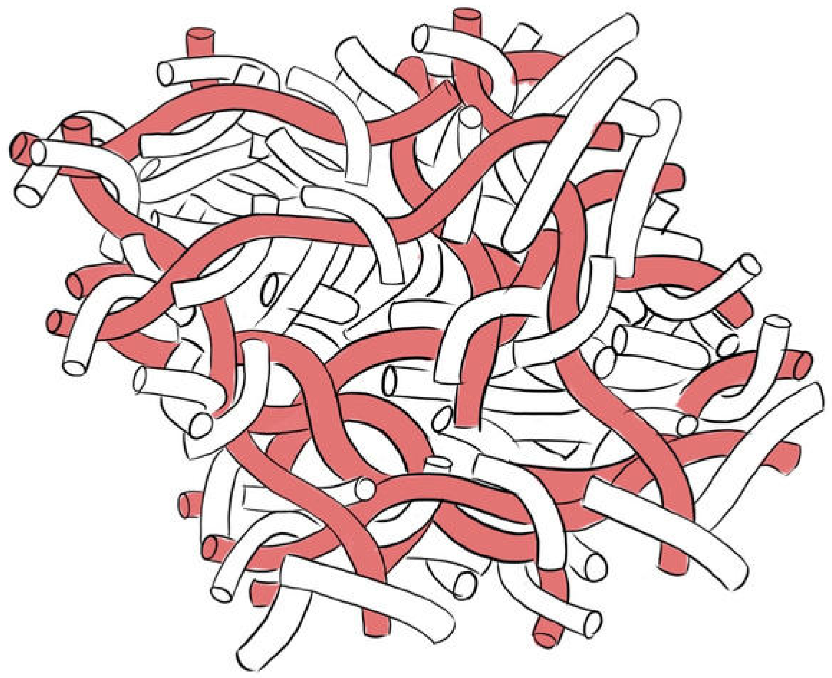
Types of (Conductive) Fabrics



1. **Woven**



1. **Knit**



1. **Non-woven**

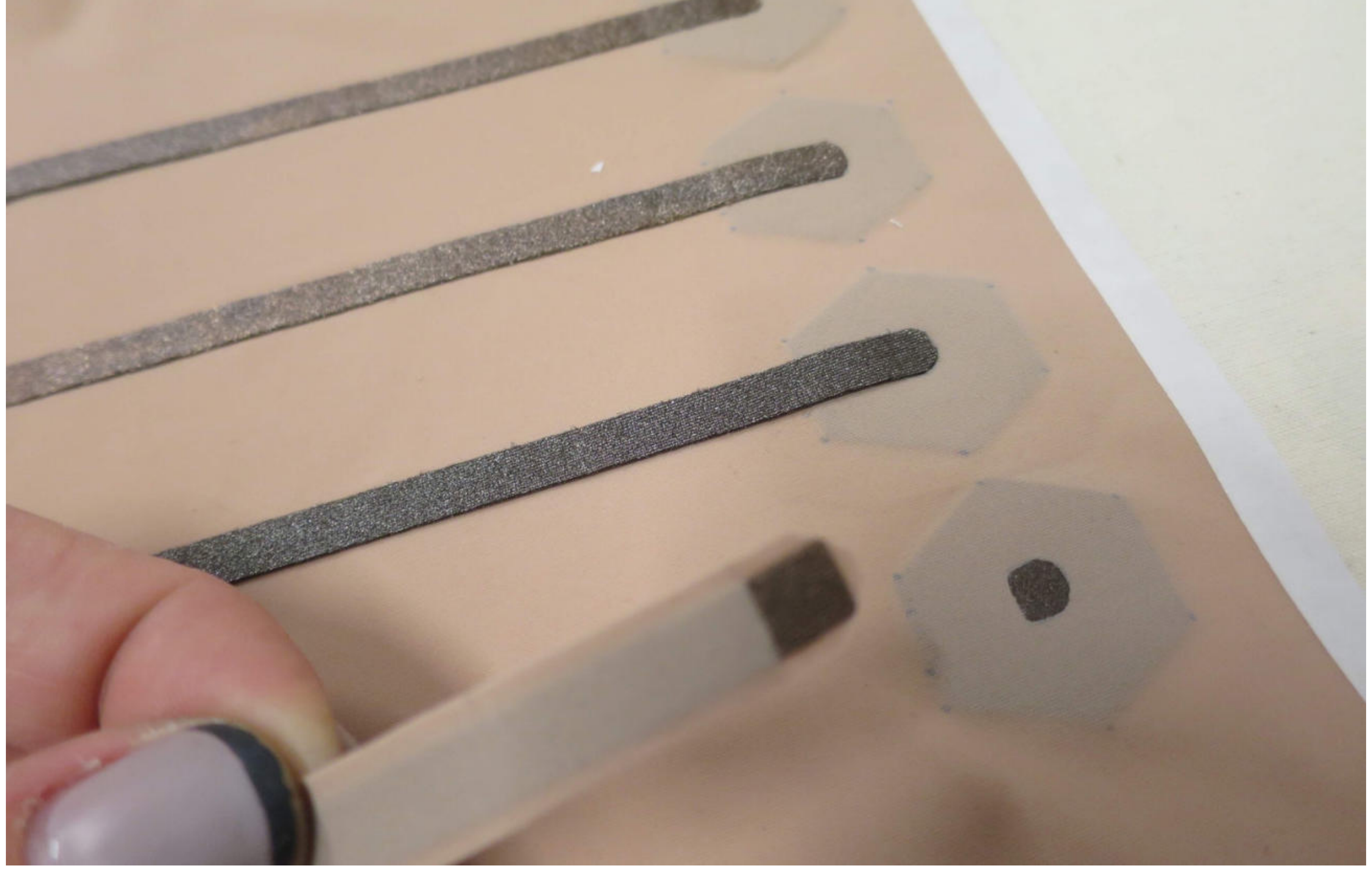
Metallized Nylon Fabric (Woven)

1. Metallized, woven textiles (e.g., nylon)
2. Different compositions of metals, e.g., copper, silver (Shieldex Kassel) or silver, copper and tin (Shieldex Zell).
3. Not stretchable
4. Sewable, can be ironed, some solderable
5. Surface resistivity (constant): <0.02 - 0.03 $\Omega/\square$, i.e., **highly conductive.**
6. **Important!** Conductive fabrics are also commonly used for shielding applications. Thus, some may contain nickel. **Do not use fabrics containing nickel** for wearable computing, especially not on skin!

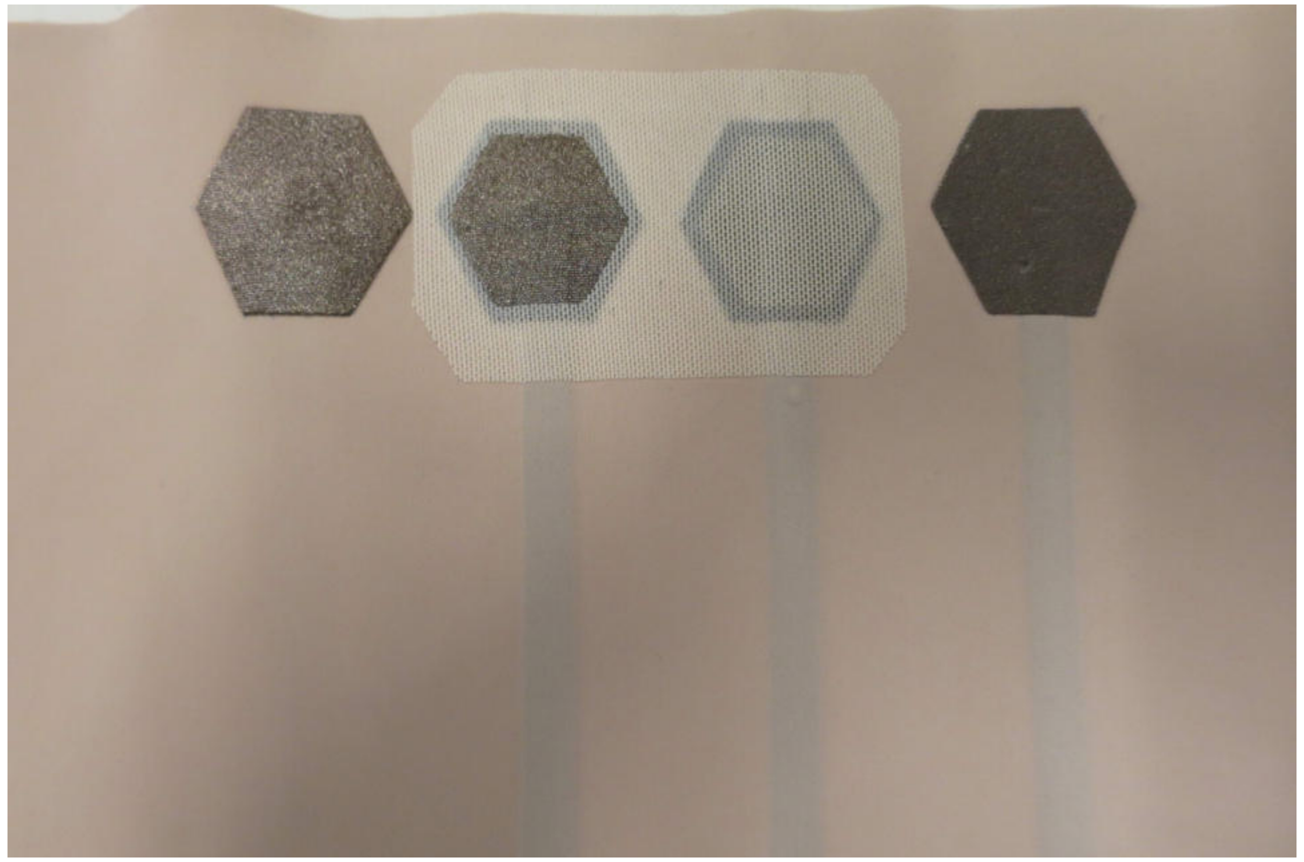
Example Material

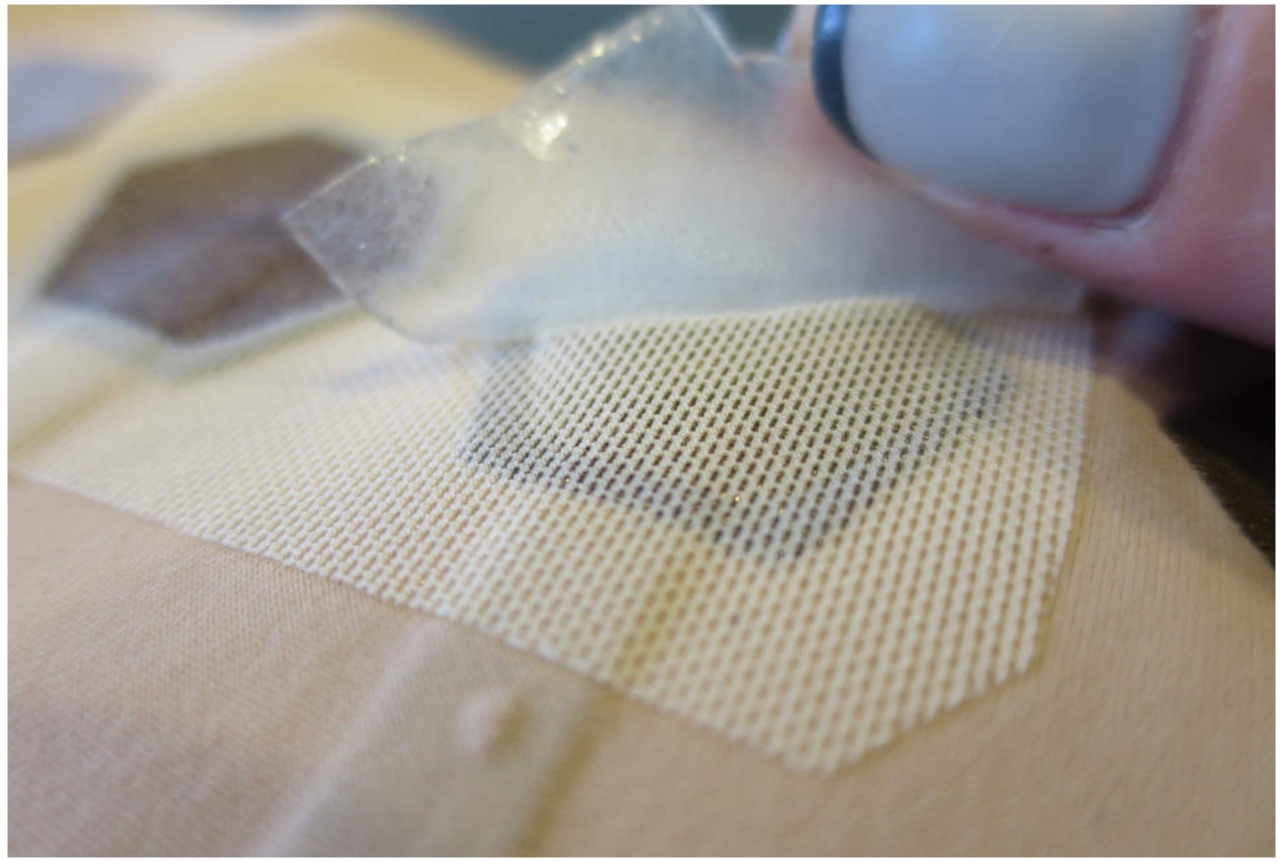
Stretchable (Knit) Conductive Fabric

1. Metallized, knitted textiles (e.g., polyamide + elastomers)
2. Different variants, e.g., uni-directional or bi-directional stretch.
3. Sheet resistivity: can be <2 Ω/□ (Technik-tex P130) to < 5 Ω/□ (Silitex), i.e., **highly conductive**, suited for on-skin electrodes.
4. Resistivity slightly variable, can be used as stretch sensors
5. Available as (e.g.,)
   a. Shieldex® Technik-tex P130
   b. Shieldex® Technik-tex P180
   c. Shieldex® Silitex

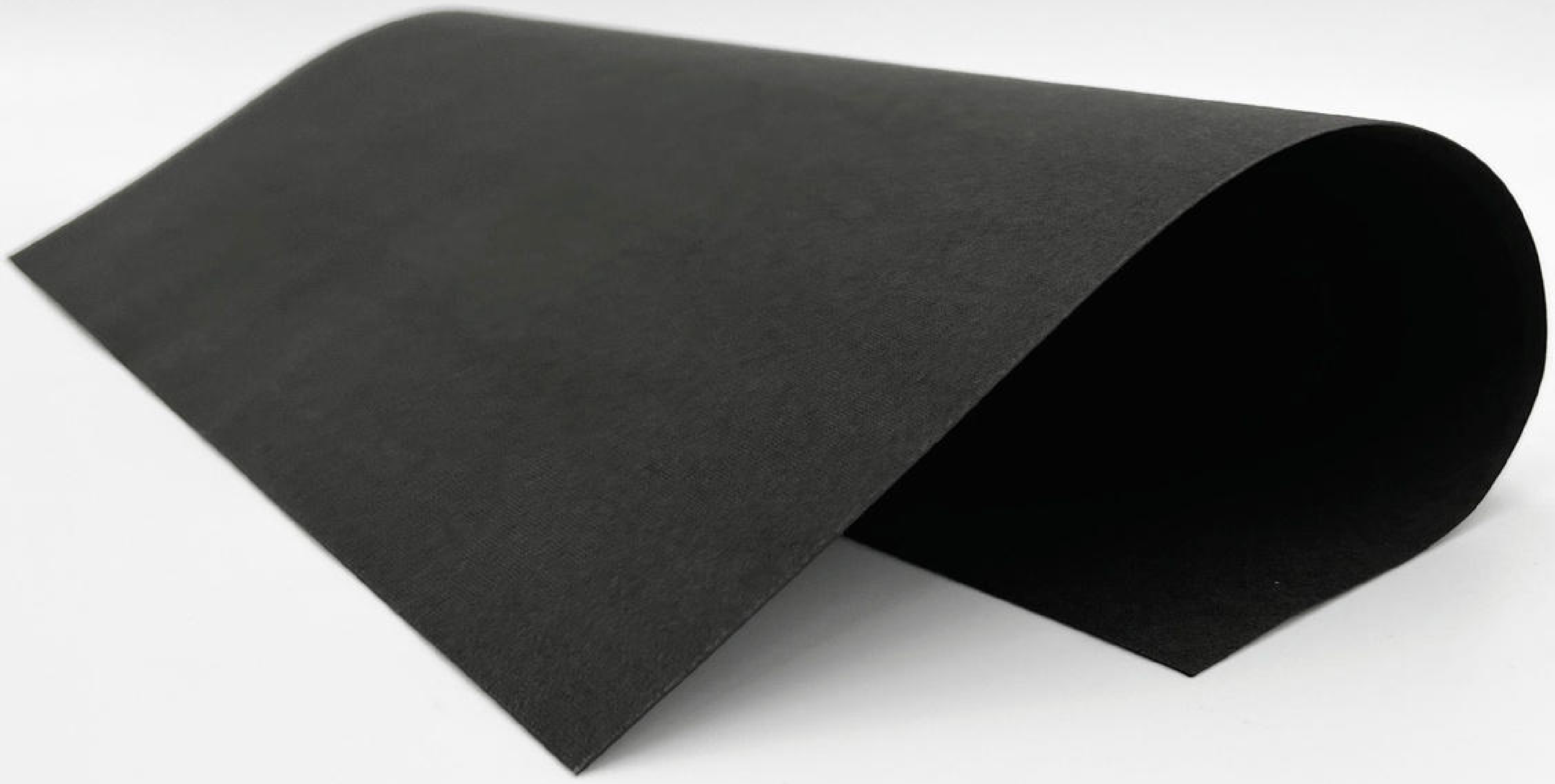


1. **Skillsleeves (TEI 2021)**
2. Jarrod Knibbe, Rachel Freire, Marion Koelle, and Paul Strohmeier





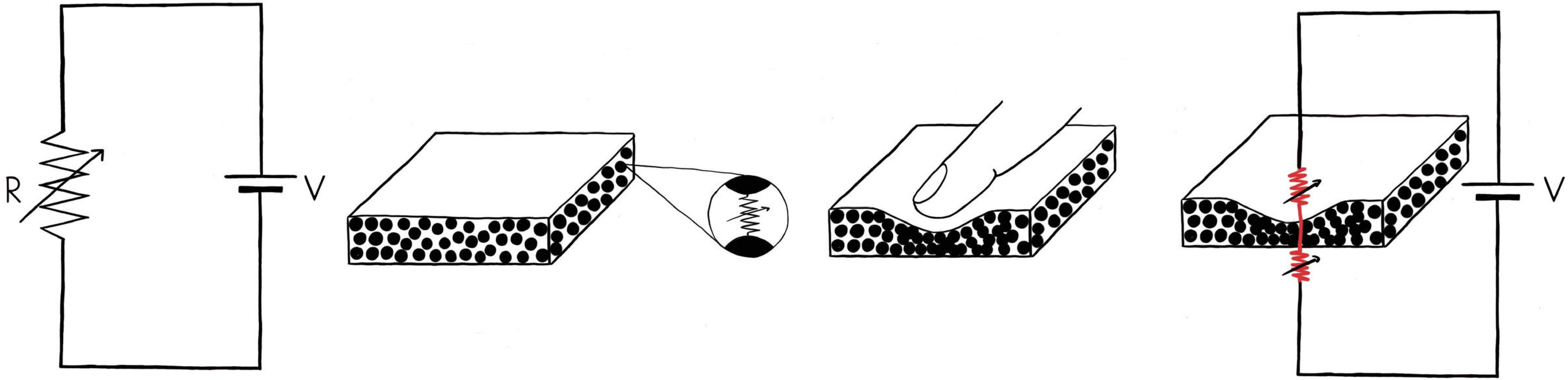
Example Material

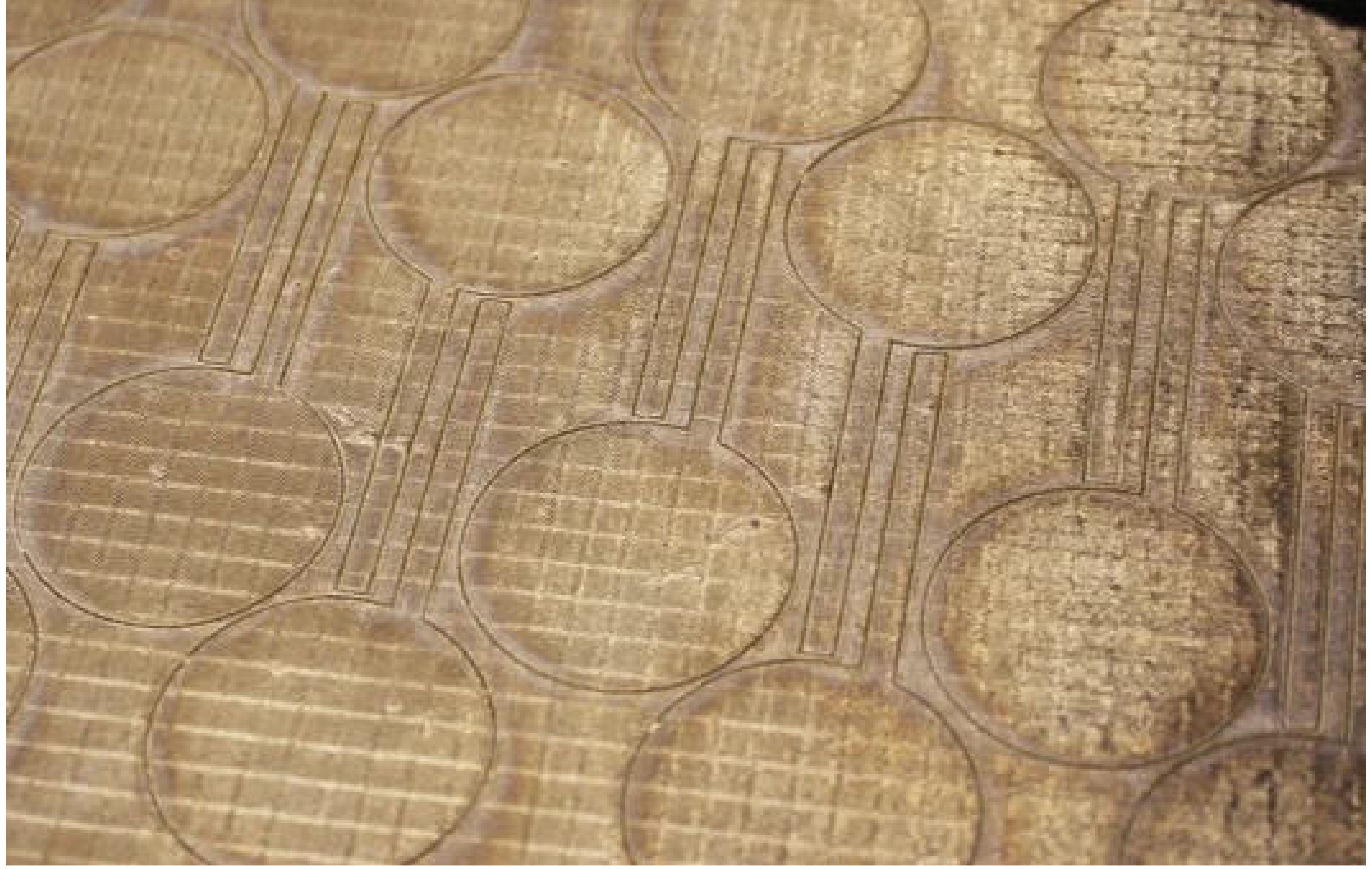


(Piezo) Resistive Fabric

1. Non-woven fiber, coated with conducted polymer (polymerized): ”carbon-doped”
2. Relatively stiff and not stretchable
3. **Variable surface resistivity** of 8 Ω/☐ (compressed) to 105 Ω/☐ (relaxed)
4. High resistivity in relaxed state, suitable as heater fabric
5. Available, e.g., from Eeonnyx:
   a. EeonTex NW-170 Pl-20
   b. https://www.adafruit.com/product/3670 (on Mouser)

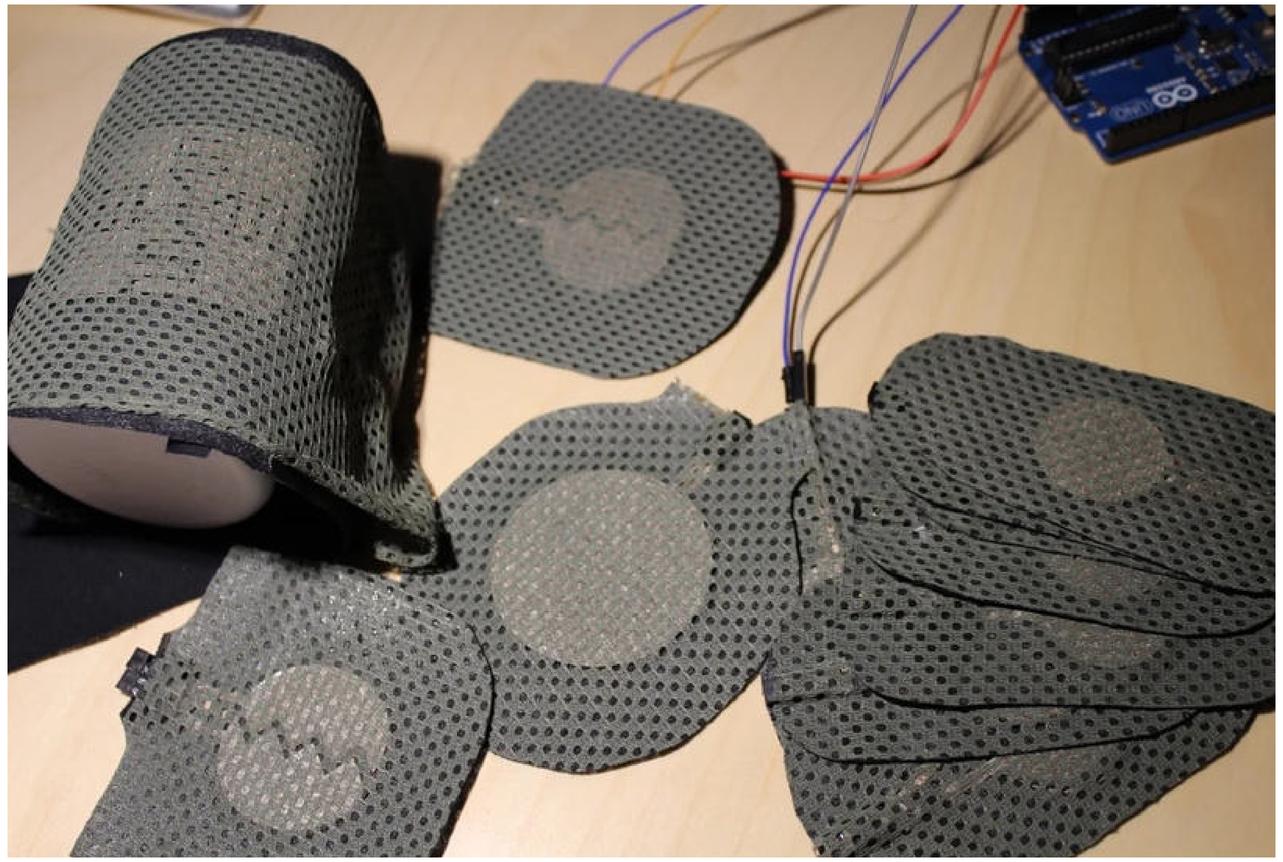
Piezoresistivity





1. **zPatch (TEI 2018)**
2. Paul Strohmeier, Jarrod Knibbe, Sebastian Boring, and Kasper Hornbæk.

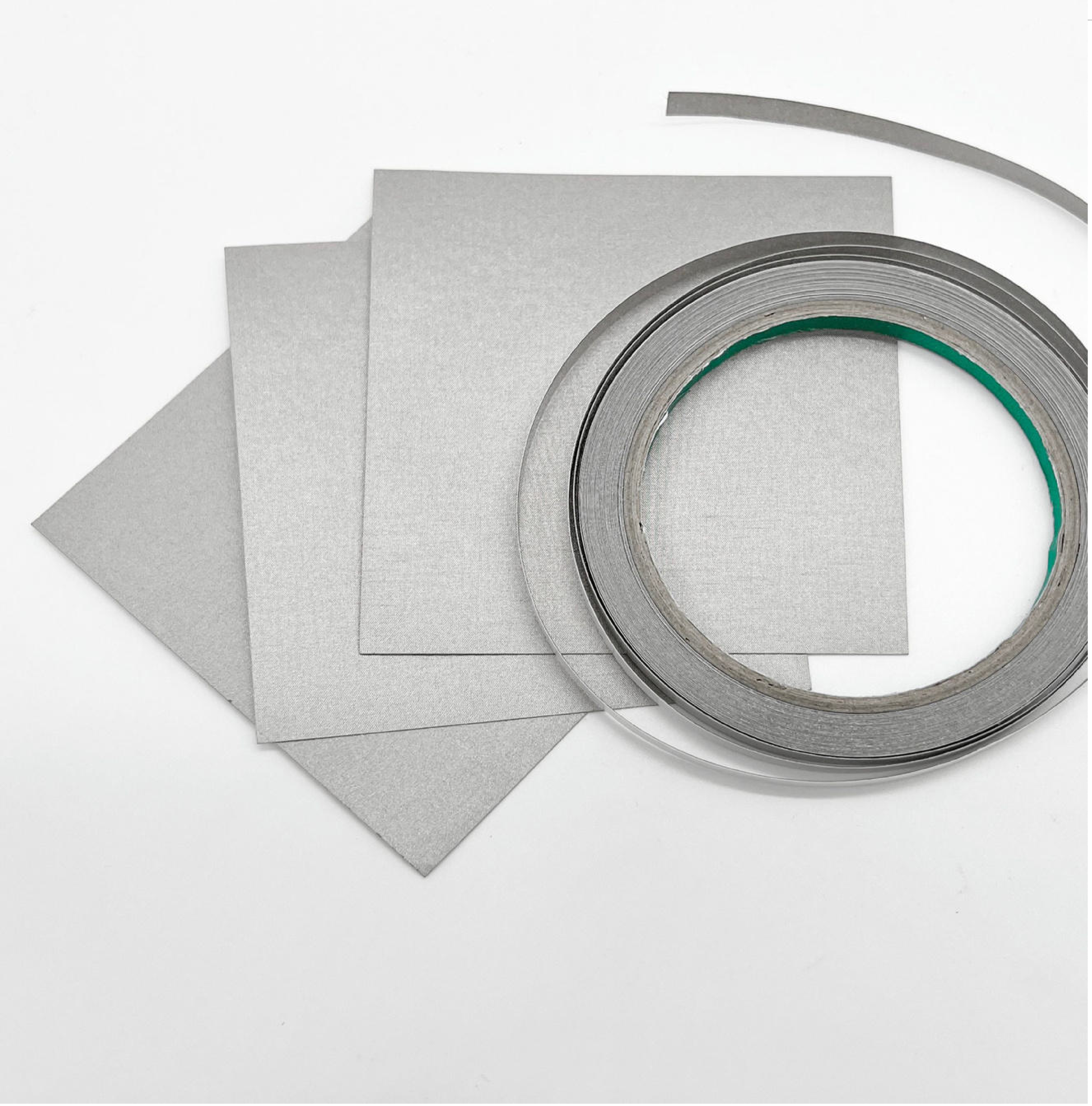
3. https://zpatch.github.io
4. https://www.instructables.com/ZPatch-Hybrid-ResistiveCapacitive-ETextile-Input/



Resistive Sheet Material

1. Polymer containing carbon particles "carbon-doped" cast into sheets
2. Usually sold as anti-static shielding material, re-purposed for e-Textile pressure sensing.
3. Relatively stiff and not stretchable
4. Heat sealable (e.g., using a soldering iron)
5. **Surface resistivity**: ca. 31 Ω/☐
6. Available, e.g., as "Velostat" or "Linqstat":
   a. https://www.adafruit.com/product/1361 (on Mouser)

Example Material

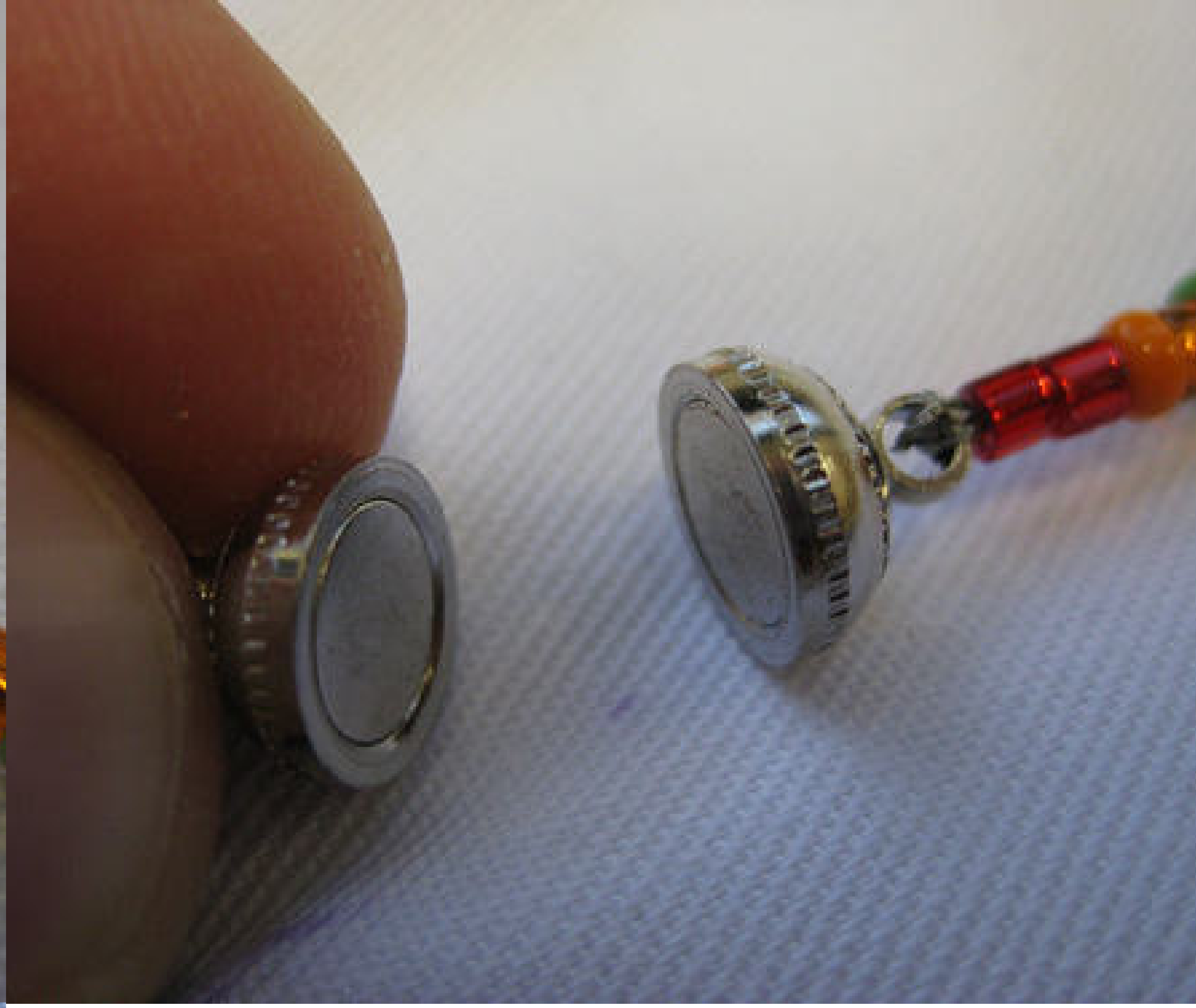
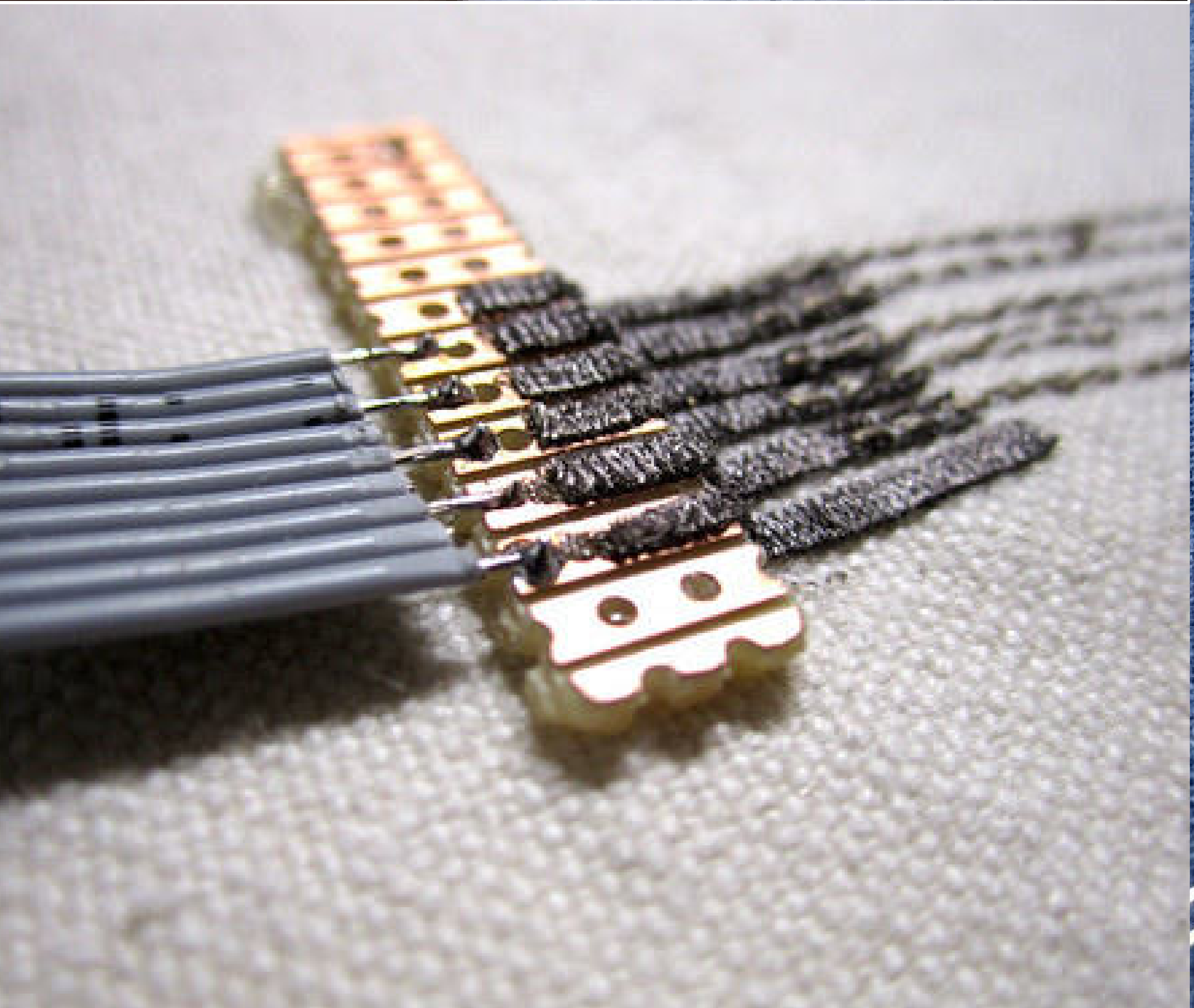
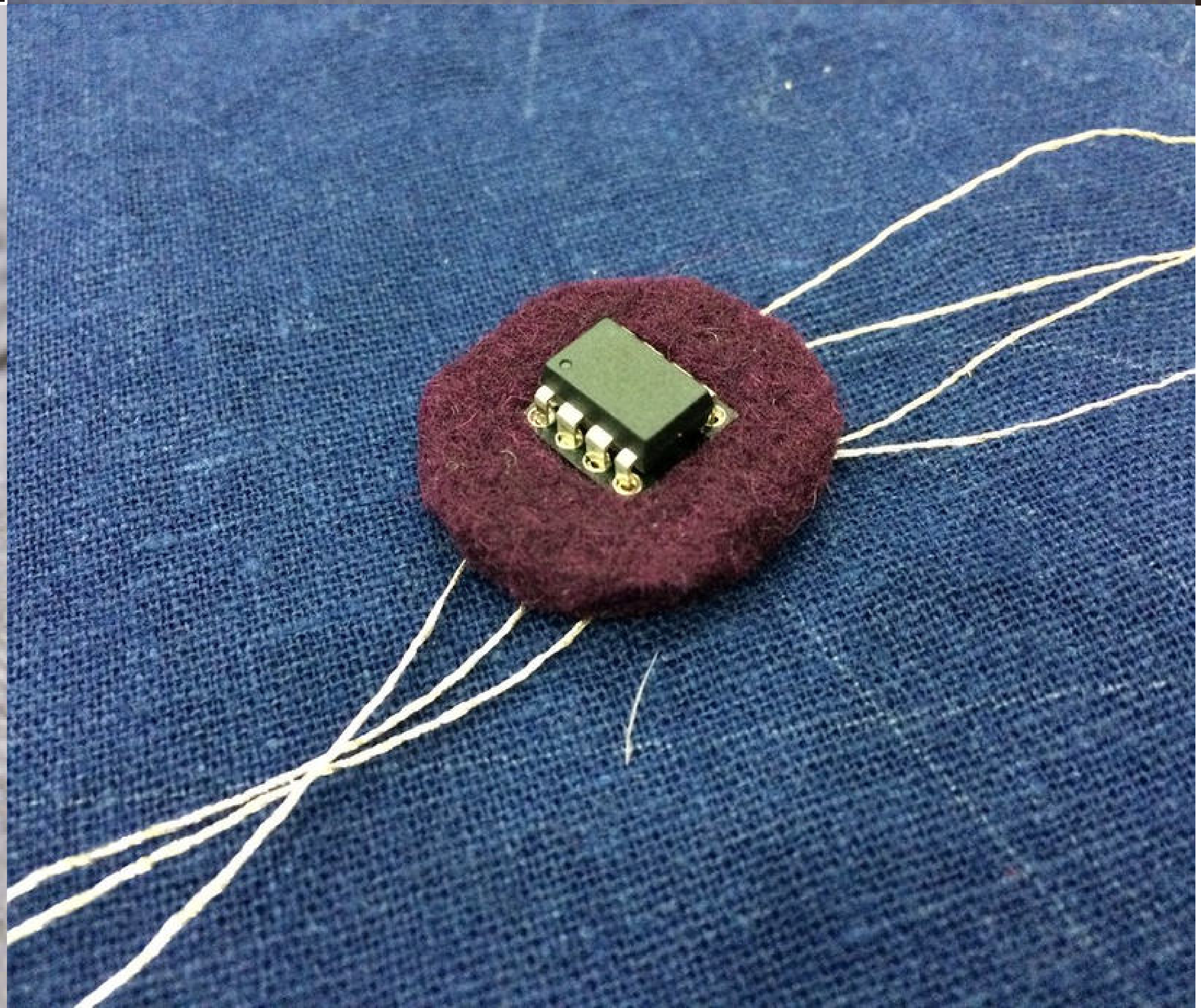


Silver-plated Nylon Adhesive Fabric

1. Metallized silver nylon fabric (or tape) with adhesive backing.
2. Relatively stiff compared to other conductive fabrics.
3. Easy to handle, easy to cut into shape.
4. Surface resistivity (constant): approx. 5-6 Ω/□., i.e., **highly conductive**
5. Solderable (with practice and flux)
6. Easily available, e.g., from Adafruit or Mouser:
   a. https://www.adafruit.com/product/4762 (on Mouser)
   b. https://www.adafruit.com/product/3960 (on Mouser)

Component Connection

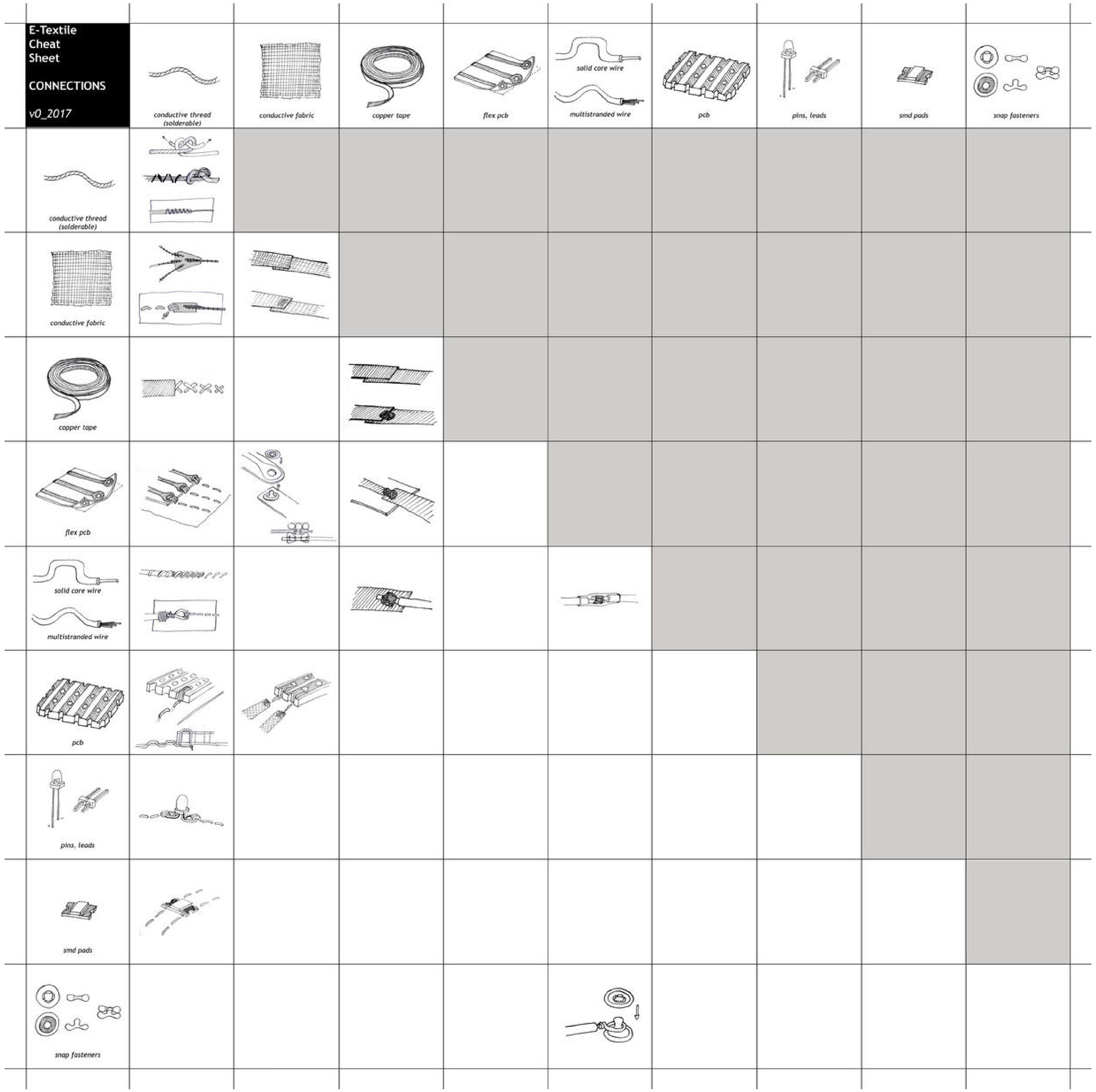


Hannah Perner-Wilson and Mika Satomi: Kobakant,
Examples online at:

- https://www.kobakant.at/DIY/?p=1272

  and
- https://www.kobakant.at/DIY/?p=7963



E-Textile Connection Cheat Sheet

1. Challenges:
2. electrical connections (i.e., „making contact“)
3. insulating overlapping traces/components (i.e., “preventing contact“)
4. different techniques, inspired from electronics and crafts > cheat sheet

1. E-Textile Cheat Sheet, http://www.kobakant.at/DIY/?p=6921
   Full resolution version by Plusea/Hannah Perner-Wilson on Flickr (https://www.flickr.com/photos/plusea/36173554336) under CC BY 2.0 DEED License

Meet the Materials | Group Work

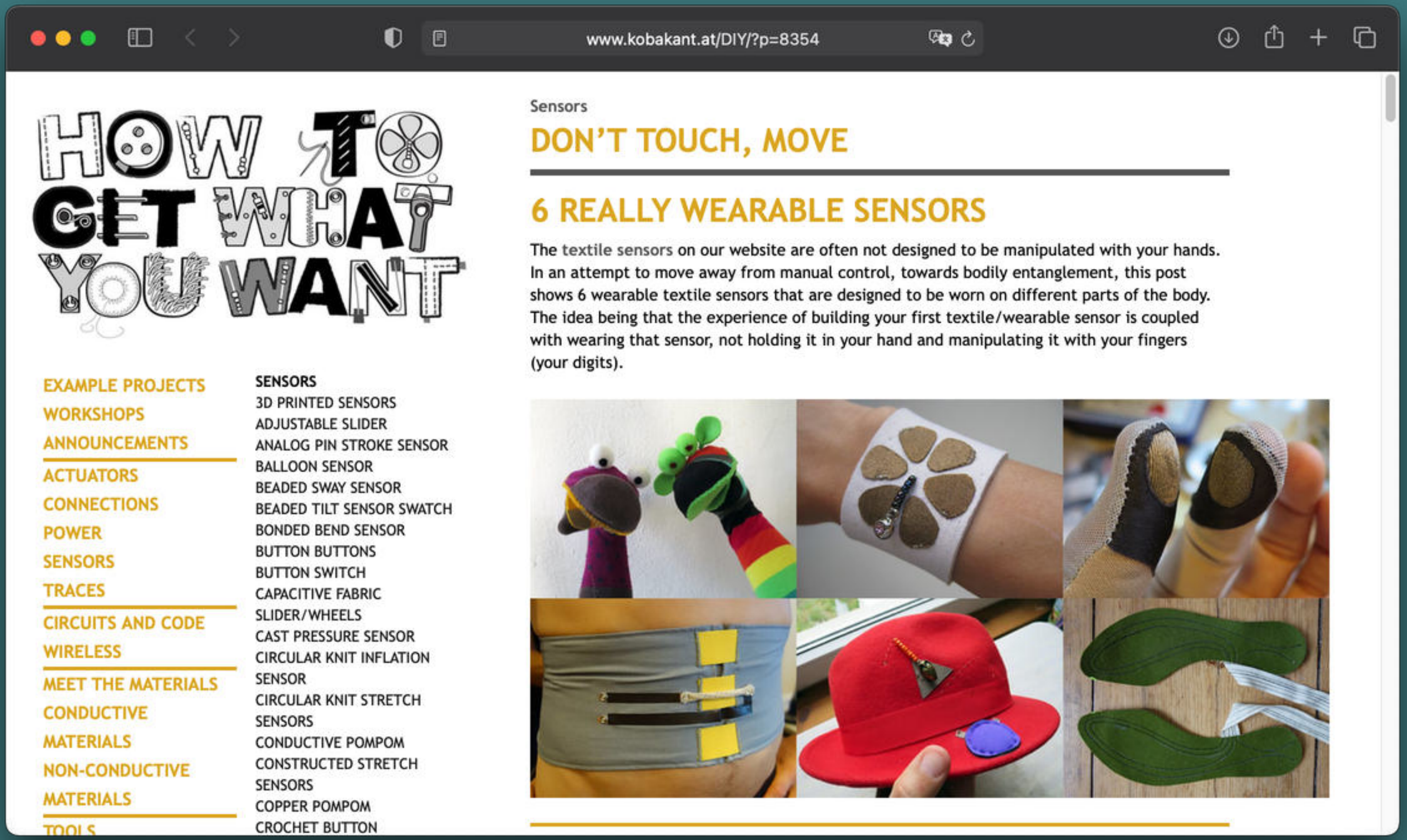
Material (e.g., yarn, fabric)	**Method & Process**	**Measured R**	**The Material's Resistivity**

Your Task: Pick a multimeter and explore the available materials.

1. Test a selection of the materials from the material collection.
2. Determine **surface resistivity** (for thin, sheet material) or **volume resistivity** (for yarns or foam material). Does your measure equal what is provided on the data sheet (if available)?
3. Prepare to **discuss your results** orally in class (max. 5 Min). Use above table as orientation.

(max. 15 Min)

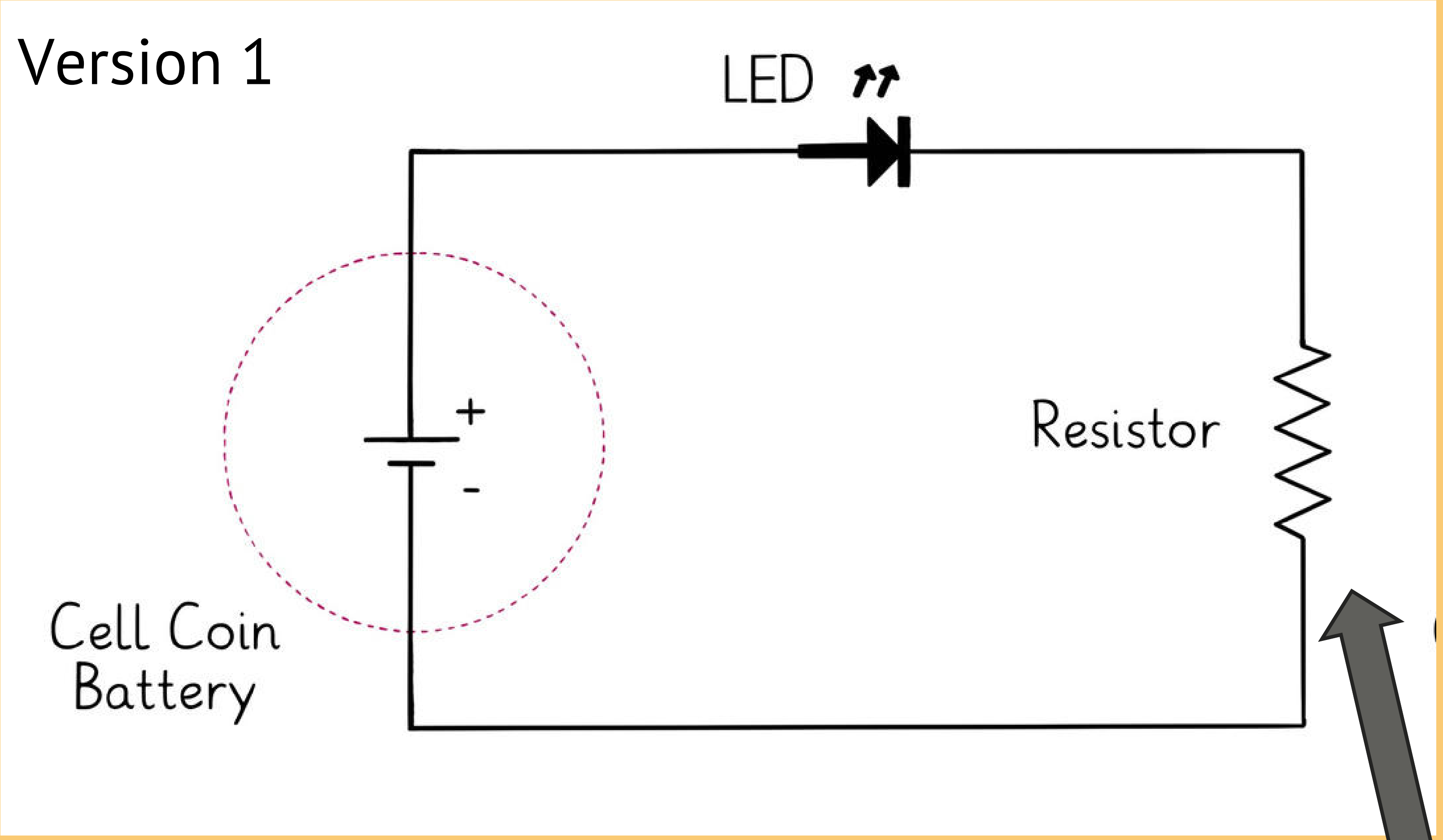
Literature / Further Reading



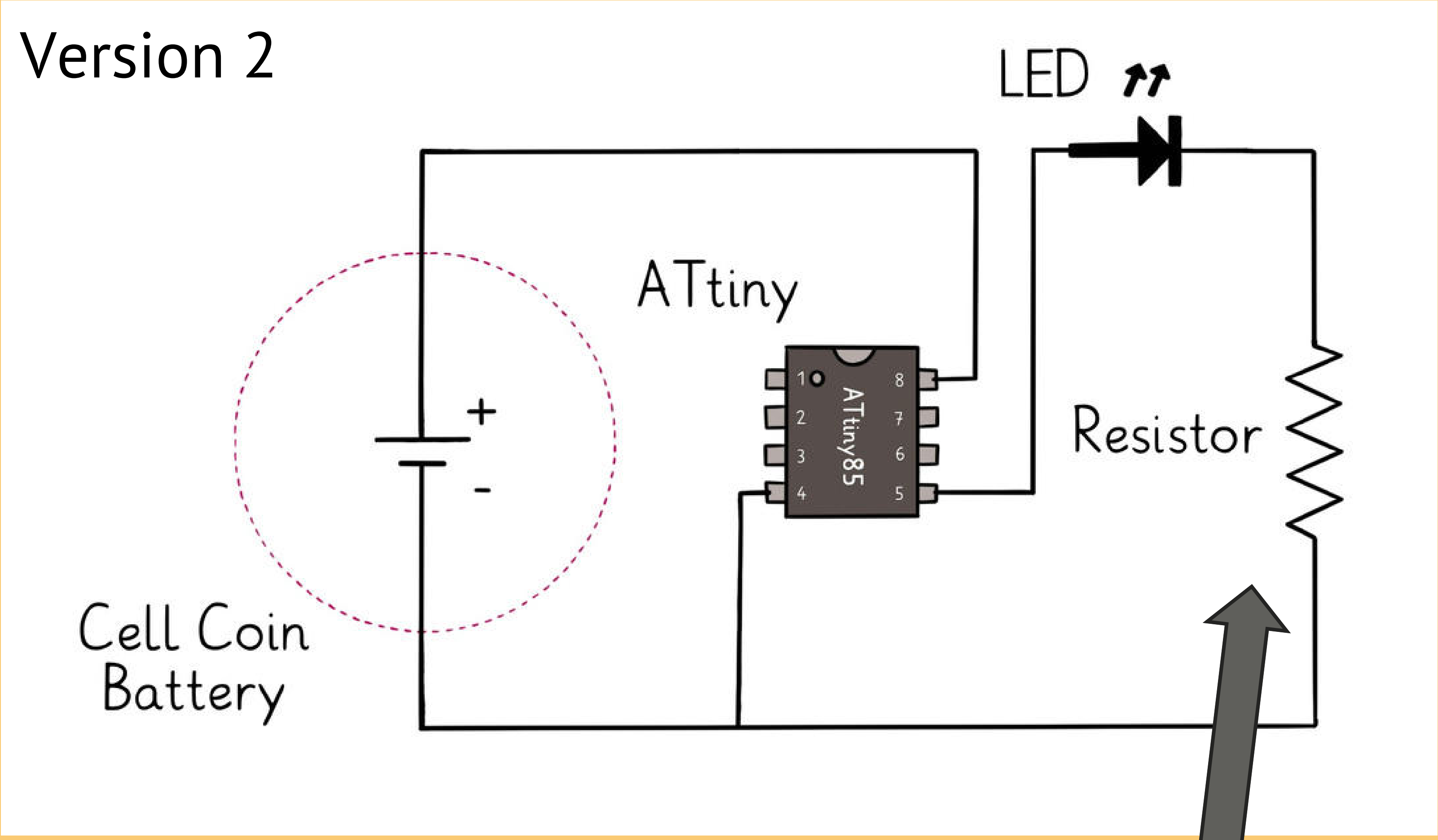
1. Perner-Wilson, H., Satomi, M.: Kobakant (www.kobakant.at), E-Textile Knowledge Repository, Material Data Base & Blog, e.g., "Bend Sensor Glove"
2. Perner-Wilson, H., Buechley, L., & Satomi, M. (2010, January). Handcrafting textile interfaces from a kit-of-no-parts. In *Proceedings of the fifth international conference on Tangible, embedded, and embodied interaction* (pp. 61-68). see also: http://konp.plusea.at
3. Buechley, L., Eisenberg, M., Catchen, J., & Crockett, A. (2008, April). The LilyPad Arduino: using computational textiles to investigate engagement, aesthetics, and diversity in computer science education. In Proceedings of the SIGCHI conference on Human factors in computing systems (pp. 423-432).

E-Textile Exercise

Version 1 is mandatory; Version 2 is a bonus exercise. Pick based on your skill level/previous knowledge and time commitment!



OR



The LEDs we're using already have a resistor built in, so they can be connected directly to the positive and negative terminals of the battery.